

Cosmological Predictions of the GTE/ Φ_{MDL} Framework: Dark Energy, the CMB Spectral Tilt, and Gravitational Signatures from First Principles

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Classification: Theoretical Physics; Cosmology; Quantum Gravity; Cellular Automata; Information Theory.

Abstract

We derive the principal cosmological observables of the Generative Triple Evolution (GTE) / Φ_{MDL} framework from zero-parameter first principles, with central algebraic steps certified in Lean 4.

The dark-energy fraction is fixed by two independent structural routes: PSC route: $\Omega_\Lambda = 0.6899 (+0.18\sigma)$ from Planck 2018; **A/D**); holographic route: $\Omega_\Lambda = 3\pi/14 = 0.6732 (-2.8\sigma)$; together these bracket Planck 2018. We prove that this bracketing is structural, not coincidental: the holographic value is the price of the MDL-minimal self-description carrier (a lower bound) and the PSC value is the capacity ceiling of the same adjudication ledger (an upper bound), with the inter-route ordering reducing to the machine-certified GTE-atom inequality $14 \ln(2000/3) > 9\pi^2$ (**A_{Lean}**). The exact realized fraction is a left-computably-enumerable real of Turing degree **0'** — definite but not computable — so no computable selection mechanism for Ω_Λ exists, and the two routes are its computable endpoints; the census endpoint becomes a 3σ -decidable test at precision $\sigma^* \approx 3.4 \times 10^{-4}$. The bracket is non-empty only for $N_{\text{gen}} \leq 3$ (measurement-free); with the PSC Layer-I bound $N_{\text{gen}} \geq 3$ this determines $N_{\text{gen}} = 3$ without Presentation Invariance (**A/D**, conditional on named structural premises). The associated energy density $\rho_\Lambda = (9/112) M_{\text{Pl}}^2 H_0^2$ agrees with observation at the few-percent level; a PSC boundary condition fixes $w = -1$. The $\mathbb{Z}_7$ vacuum is anomaly-free (machine-certified, zero **sorry**), and holographic mode counting ($3L$ rather than L^3) suppresses one-loop vacuum corrections by $\approx 10^{-42}$.

A new defect-cosmology sector is derived: the $\mathbb{Z}_7$ order, restored after reheating, re-orders at $T_G \approx 0.70$ GeV and forms a transient Kibble domain-wall foam; the canonical coupling $\varepsilon\varphi^2(D_\mu\chi)^2$ ($\varepsilon = 7/9$, **A_{Lean}**) lifts the vacuum-label degeneracy and collapses the foam within $\sim 10^{-23}$ s — 700× before BBN — selecting the vacuum $k = 0$ and leaving zero surviving defect network ($\Omega_{\text{GW}} h^2 \lesssim 3 \times 10^{-47}$ at $f \approx 0.02$ Hz; $\Delta N_{\text{eff}} \approx 0$), a parameter-free falsifiable prediction.

The baryon-to-photon ratio is derived via the Φ_{MDL} kink-top coupling (FKTT): $\eta_B = 6.109 \times 10^{-10}$ ($+0.15\sigma$ from Planck 2018/BBN $(6.10 \pm 0.06) \times 10^{-10}$; **A_{Lean}**, zero **sorry**, zero axioms).

Further predictions from the same integer inputs include: the CMB spectral index $n_s = 1 - \ln 2/(2\pi^2) = 0.96488$ (-0.005σ , $\mathbf{A}_{\text{Lean}}$, fourteen Lean theorems); the tensor-to-scalar ratio $r = 0$ exactly (no primordial gravitational waves; primary LiteBIRD falsifiable prediction); the neutrino mass sum $\Sigma m_\nu = 59.4$ meV; the CKM CP phase $\delta_{CP} = 68.51^\circ$ (0.017% from PDG); and the Hubble constant $H_0 = 67.95$ km/s/Mpc ($\mathbf{A}$ with $T_{\text{CMB}} = 2.7255$ K as the sole external input; $\mathbf{A/D}$ fully first-principles via Path D, `gtb_kink_overlap_certified`). An explicit falsifiability profile is provided.

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Claim-strength taxonomy

Every result in this paper is tagged with its confidence level:

- **A_{Lean}** — machine-certified in Lean 4 with zero **sorry**. *Conditional* **A_{Lean}** uses a named physics axiom, always stated.
- **A_{MDL}** — minimum-description-length-unique within a declared expression class.
- **A/D** — full analytic derivation; computationally confirmed.
- **A** — computationally verified (numerical script); analytic derivation or Lean certificate remains open.
- **B** — plausible and partially supported.
- **D** — exploratory, long-term open, or conjectural.

Ordering: **A_{Lean}** > **A_{MDL}** > **A/D** > **A** > **B** > **D**. Lean theorem names appear in **monospace**; all cited theorems reside in the canonical **ugp-lean** library.

What this paper claims

Analytically derived and machine-certified (A/D/A_{Lean}):

- **[Dark energy, route 1]** $\Omega_\Lambda = (\ln 2/3\pi) \log_2(2000/3) = 0.6899$, $+0.18\sigma$ from Planck 2018, zero free parameters (**omega_lambda_from_d_res**, **psp_epoch_selection_master**; **A/D**).
- **[Dark energy, route 2]** $\Omega_\Lambda = \tau \cdot (\pi/2) = 3\pi/14 = 0.6732$ from the three-tape holographic mode count, with $\tau = 3/7$ derived from Rule-110 ether dynamics (**omega_lambda_from_temporal_voxel**, **tau_three_sevenths_from_ether**; **A/D**).
- **[Two-sided structural bracket]** The two routes are the two endpoints of one object: the holographic value prices the MDL-minimal self-description carrier (lower bound) and the PSC value is the capacity ceiling of the same adjudication ledger (upper bound); the inter-route ordering is the GTE-atom inequality $14 \ln(2000/3) > 9\pi^2$. (**cc_floor_orientation_atom_inequality**, **omega_lambda_bracket_strict**; **A_{Lean}**; bracket theorem **A/D**, conditional on named structural premises, [Theorem 3.1](#)).
- **[No computable selection]** The realized Ω_Λ is a left-c.e. real of Turing degree **0'** ($\Delta_2^0 \setminus \Delta_1^0$): definite but not computable; no computable selection mechanism or convergence modulus exists; census endpoint decidable at 3σ at precision $\sigma^* \approx 3.4 \times 10^{-4}$. (**no_computable_gte_rt_convergence_modulus**, **sigma_star_falsifiability_horizon**; **A_{Lean}**; classification **A/D** conditional).
- **[Measurement-free generation-count exclusion]** The admissible interval $[\text{floor}(N), \text{census}(N)]$ is non-empty only for $N \leq 3$; the continuous boundary is $N^* = 3.037$; with the PSC Layer-I bound $N_{\text{gen}} \geq 3$ this pincers $N_{\text{gen}} = 3$ without Presentation Invariance. (**ngen_bracket_orientation_flip**, **ngen_pincer_from_layer1_and_orientation**; **A_{Lean}**; **A/D** conditional as above).
- **[Defect cosmology]** The $\mathbb{Z}_7$ ordering crossover at $T_G \approx 0.70$ GeV forms a Kibble wall foam that the canonical coupling bias $|\Delta V| \approx 3.8 \times 10^{-2}$ GeV⁴ collapses within $\sim 10^{-23}$ s (700× before BBN), selecting the vacuum $k^* = 0$ (**A/D**, **ROBUST**) and predicting zero surviving defect network, $\Omega_{\text{GW}} h^2 \lesssim 3 \times 10^{-47}$ at $f \approx 0.02$ Hz, and $\Delta N_{\text{eff}} \approx 0$. (**vcoupling_breaks_z7_shift**, **wall_bias_minimum_unique**, **compact_completion_minimum_at_k0**; **A_{Lean}** cores).
- **[Energy density]** $\rho_\Lambda = (9/112) M_{\text{Pl}}^2 H_0^2 = N_{\text{spatial}}^2 / (D^2 |\mathbb{Z}_7|) M_{\text{Pl}}^2 H_0^2$ (**voxel_cc_coefficient**; **A/D**).
- **[Quantum protection]** Holographic one-loop/tree ratio 1.22×10^{-42} versus the standard 1.66×10^{40} ; suppression $3H_0^2/M_{\text{kink}}^2 \approx 7.4 \times 10^{-83}$ (**holographic_voxel_scaling**; **A/D**-partial).
- **[CMB tilt]** $n_s = 1 - \ln 2/(2\pi^2) = 0.96488$, -0.005σ from Planck 2018, fourteen zero-sorry theorems (**n_s_formula**, **n_s_less_than_one**, **beta_g_from_classical_rg**; **A_{Lean}**).
- **[Classical vacuum]** $V(\Phi_k) = 0$ at all kink minima; the $\mathbb{Z}_7$ vacuum energy is mass-independent (**phimdl_tmunu_vacuum_zero**, **z7_vacuum_energy_mass_independent**; **A_{Lean}**).

- **[Newton constant]** Gorard discrete-smooth curvature chain with $\kappa_{SD} = 10/13$ and $C_{Gorard} = 3/32 = N_{\text{spatial}}/(2D^2)$ (`kappa_SD_eq_10_13`, `gorard_chain_catAL_master_bundle`; $\mathbf{A}_{\text{Lean}}$).
- **[Baryon asymmetry]** $\eta_B = 6.109 \times 10^{-10}$ ($+0.15\sigma$ from Planck 2018 CMB+BBN $(6.10 \pm 0.06) \times 10^{-10}$; $\mathbf{A}_{\text{Lean}}$, zero sorry, zero axioms), via the FKTT mechanism (Φ_{MDL} Kink–Top Coupling; `kink_top_coupling_eq_eps_FN`, `phi_md1_kink_bps_saturation`; $\mathbf{A}_{\text{Lean}}$). The asymptotic sech formula gives $\eta_B = 6.36 \times 10^{-10}$ ($+4.3\sigma$) as an upper bound (`yukawa_overlap_exponent_catad`, `leptogenesis_kink_overlap_catad`, `d_top_derivation_chain_catad`; $\mathbf{A}_{\text{Lean}}$ conditional).
- **[Hawking kink emission]** BPS critical mass $M_{\text{kink}}^{\text{crit}} = (8/49)m_\tau = 290.10$ MeV; Bose-enhanced emission profile (`kink_crit_mass_formula`, `hawking_kink_emission_catad`; $\mathbf{A}/\mathbf{D}$).
- **[Strong-field UV bound]** $V_{\text{max}} = 2m^2/49$ (`strong_field_uv_bound_catad`; $\mathbf{A}/\mathbf{D}$).
- **[CKM CP phase]** $\delta_{CP} = \pi/2 - 3/8 = 68.51^\circ$, 0.017% from PDG (`ckm_cp_phase_formula`; $\mathbf{A}$, Lean-certified identity).

Computationally verified ($\mathbf{A}$, reproducible):

- Neutrino sum $\Sigma m_\nu = 59.4$ meV and lightest mass $m_{\nu_1} = 0.679$ meV; leptogenesis scale $M_{R,1} = 1.1105 \times 10^{13}$ GeV, GTE-structurally derived via three-tape BPS instanton suppression ($\mathbf{A}$; eq. (29)).
- CKM Jarlskog invariant $J = 3.02 \times 10^{-5}$ (5% from PDG); leptogenesis efficiency factor $K_1 = 15.93$ (feasible).
- Leptonic (PMNS) Jarlskog invariant $J_{\text{GTE}}^{(\text{PMNS})} = -(8184\sqrt{437}/5057221)\sin(\pi/7) \approx -1.468 \times 10^{-2}$ (`gte_jarlskog`, `gte_jarlskog_is_negative`; $\mathbf{A}_{\text{Lean}}$, zero sorry).
- χ_1 direct detection cross-section $\sigma_{\text{SI}}(\chi_1-N) = 9.76 \times 10^{-61} \text{ cm}^2$ ($\mathbf{A}/\mathbf{D}$); 28–29 orders below CRESST-III; χ_1 is ultra-feeble dark matter (UFIDM).

What this paper does NOT claim

- That the quantum cosmological-constant hierarchy is fully solved. The holographic non-renormalization result is $\mathbf{A}/\mathbf{D}$ -partial: the phenomenological question “why do loops not overwhelm the constant?” is answered, but the complete Φ_{MDL} path-integral derivation of the $3L$ (not L^3) mode count is a future result.
- That H_0 is derived purely from first principles with no observational input. The partial derivation (Path D, §6.3) requires $T_{\text{CMB}} = 2.7255$ K as the sole external input and achieves $\mathbf{A}$ status (computationally verified): $H_0 = 67.95$ km/s/Mpc derived non-circularly. The fully first-principles Path D chain (T_{CMB} derived from the η_B chain, no external input) achieves $\mathbf{A}/\mathbf{D}$ status (`gtb_kink_overlap_certified`, $\mathbf{A}_{\text{Lean}}$). The sole remaining open piece is deriving T_{CMB} from the reheating temperature ($\mathbf{D}$).
- That the CMB tilt is fully derived at the level of field theory: the algebraic chain is $\mathbf{A}_{\text{Lean}}$, but the physical identification of the binary ($\mathbb{Z}_2$) effective field theory with a classical scalar on a compact S^4 is $\mathbf{A}/\mathbf{D}$ and not yet $\mathbf{A}_{\text{Lean}}$.
- That the cellular automaton is the physical substrate. The CMCA is the Level-1 algebraic certificate; the Level-2 physical substrate is the Φ_{MDL} field (see [14]).
- That the two-sided bracket theorem and the generation-count pincer are unconditional. They are conditional on named structural premises (record-measure monotonicity and additivity; positive witness costs) and, for the $N \neq 3$ boundary value, on the structural ansatz for the N -dependence of the two formulas; the absolute per-mode pricing of the carrier is identified by cross-sector consistency, not derived ab initio (Section 3.5).
- That a confirmed cosmological domain-wall relic would falsify the whole framework. It would falsify the defect-cosmology chain — the coupling-bias annihilation mechanism and the $\mathbb{Z}_7$ ordering-crossover epoch — not the substrate identification or the particle-spectrum derivations (Section 7).

1 Introduction

This paper is part of the UGP Physics research programme — the quantitative branch of the broader Reflexive Reality project [15, 25] — which derives Standard Model and cosmological parameters from a single combinatorial substrate: the Generative Triple Evolution (GTE) polynomial over $\text{GF}(7)$. The GTE framework has, over a sequence of machine-certified papers, established the chiral Minkowski spacetime structure (P41 [27]), the continuum quantum field (P42–P43 [14, 20]), emergent gravity (P38 [16]), the functional completeness of the quantum gravity sector (P44 [24]), and the three-tape 3+1D construction (P45 [28]). Each paper derives a quantitative prediction from the same zero-free-parameter substrate.

This paper presents the first complete GTE treatment of cosmological observables, with several genuinely new results that have not appeared in any prior programme paper. The centrepiece is a three-level resolution of the cosmological constant problem: (i) the CC *magnitude* is derived for the first time from the CMCA holographic architecture, yielding $\rho_\Lambda = (9/112) M_{\text{Pl}}^2 H_0^2$ from purely structural GTE constants; (ii) the CC *value* is fixed by two structurally independent routes, $\Omega_\Lambda = 3\pi/14$ (holographic) and $\Omega_\Lambda = 0.6899$ (PSC epoch-selection), which bracket the Planck observation of 0.6889 from above and below; and (iii) quantum loop corrections are shown for the first time to be suppressed by $3H_0^2/M_{\text{kink}}^2 \approx 7.4 \times 10^{-83}$ relative to the classical CC, resolving the hierarchy without supersymmetry.

The two-route bracketing is moreover proved to be a structural theorem rather than a numerical coincidence: the holographic value is the price of the MDL-minimal self-description carrier, the PSC value is the capacity ceiling of the same adjudication ledger, the realized value is a definite but non-computable real of Turing degree $\mathbf{0}'$ lying strictly between them, and the bracket is non-empty only for $N_{\text{gen}} \leq 3$ — a measurement-free upper bound on the generation count (Section 3.5).

A further new result is the complete defect-cosmology sector (Section 7): the $\mathbb{Z}_7$ domain-wall foam that forms at the ordering crossover $T_G \approx 0.70$ GeV is annihilated by the canonical coupling bias within $\sim 10^{-23}$ s, selecting the vacuum $k = 0$ and yielding the parameter-free prediction of zero surviving defect network. Additionally, we report the first GTE derivation of the CKM CP phase $\delta_{\text{CP}} = \pi/2 - 3/8 = 68.51^\circ$, agreeing with the PDG value to 0.017%.

Cosmological predictions that appeared as subsections in prior papers are also reassembled here with a unified derivation, a complete Lean 4 certification inventory, and a single comparison table against observation. The remainder of this introduction states the cosmological constant problem, the two-level architecture that organizes every claim, and the relation of this paper to the rest of the programme; a roadmap of the paper and an overview table of all derived observables are given in Section 1.4.

1.1 The cosmological constant problem

The cosmological constant problem is the largest quantitative discrepancy in physics. A naive sum of quantum vacuum-energy contributions up to the Planck scale exceeds the observed dark-energy density by roughly 120 orders of magnitude [31]. The standard responses fall into three families: fine-tuning of a bare constant against vacuum contributions to one part in 10^{120} ; an anthropic selection over a landscape of vacua [30]; or a dynamical relaxation mechanism, of which Weinberg’s no-go theorem shows that no scalar field coupled to gravity through a local effective action can cancel the constant without reintroducing a fine-tuning of equal severity. None of these predicts the *value* of the observed dark-energy fraction $\Omega_\Lambda = 0.6889$ [1].

Holographic approaches [4, 9] reduce the discrepancy by correlating the ultraviolet and infrared cutoffs of an effective field theory, yielding the parametric scaling $\rho_\Lambda \sim M_{\text{Pl}}^2 H_0^2$, which is of the right

order. They fail, however, on the equation of state: the Cohen–Kaplan–Nelson (CKN) construction and its refinements predict $w \equiv p/\rho = 0$ over cosmological distances [9], in sharp conflict with the observed accelerated expansion, which requires $w \approx -1$.

This paper derives both the magnitude and the value of Ω_Λ from a discrete substrate with no free parameters, and shows that the equation of state is fixed at $w = -1$ by a structural boundary condition rather than by dynamics — evading both Weinberg’s no-go theorem and the CKN obstruction.

1.2 The two-level architecture

All physical claims in this paper are organized by a two-level architecture established in the GTE/ Φ_{MDL} programme [14, 20, 27].

Level 1 is the Chiral Minkowski Cellular Automaton (CMCA): a discrete algebraic certificate whose update rule is the unique minimum-description-length (MDL) polynomial $p(L, C, R) = C + R - CR - LCR$ over the finite field $\text{GF}(7)$. This is the GTE polynomial: a three-input (L, C, R) Boolean-like rule over the seven-element field that is the unique Turing-universal, MDL-minimal rule with a $\mathbb{Z}_7$ winding-number structure. The CMCA encodes the structural constraints of Standard-Model physics — the $\mathbb{Z}_7$ winding sectors (the seven residue classes mod 7 that encode the SM generation and charge structure), the generation orbit, the holographic state count $|\mathcal{S}| = 7^{3L}$ — but it is *not* the physical substrate.

Level 2 is the Φ_{MDL} field: a $\mathbb{Z}_7$ -symmetric Klein–Gordon field that is the actual continuum physical substrate. The continuum limit of the CMCA together with the Algebraic Lifting Theorem maps Level-1 structural counts onto Level-2 physical quantities [14, 20]. No finite cellular automaton can be the substrate: this impossibility is itself machine-certified [14].

Every cosmological observable in this paper is a Level-2 quantity whose value is fixed by a Level-1 count lifted through this chain. The cosmological constant is the flagship example: its observed fraction Ω_Λ is fixed independently by a $\mathbb{Z}_7$ orbit-class count and by a three-tape holographic mode count, both of which bracket the Planck value.

1.3 Why a dedicated cosmology paper

The GTE/ Φ_{MDL} programme has assembled the following prerequisites in prior papers.

- **P35** [18]: GTE unification; $\mathbb{Z}_7$ orbit structure, gauge spectrum $L_{\text{model}} = \log_2(2000/3)$, Standard Model coupling constants.
- **P38** [16]: emergent gravity from Φ_{MDL} ; the Φ_{MDL} field equations and energy–momentum tensor.
- **P41** [27]: three-layer Chiral Minkowski CMCA; spacetime structure, Lorentz invariance, Born rule, $\mathbb{Z}_7$ superselection.
- **P42** [20]: Φ_{MDL} field theory; the continuum limit, classical vacuum $V(\Phi_k) = 0$, and the Algebraic Lifting Theorem (Level-1 $\rightarrow$ Level-2).
- **P43** [14]: PSC completeness and the impossibility of a finite cellular-automaton physical substrate; the PSC non-effective self-reference mechanism (transputation).
- **P44** [24]: quantum gravity sector; the Gorard discrete-to-smooth bridge, Hawking kink emission, the $w = -1$ boundary condition.
- **P45** [28]: three-tape 3+1D construction; the shared outer clock; holographic state count $|\mathcal{S}| = 7^{3L}$; $\tau = 3/7$ from Rule-110 ether; $C_{\text{Gorard}} = 3/32$.
- **P21** [22]: neutrino mass spectrum; seesaw parameters; leptogenesis.
- **P29** [19]: dark sector; mirror-braid candidate.

- **P32** [12]: CKM matrix and CP-violation structure.

Beyond assembling cosmological predictions scattered across earlier papers, this paper contributes several results that have not appeared anywhere in the prior programme: the first derivation of the CC magnitude $\rho_\Lambda = (9/112) M_{\text{Pl}}^2 H_0^2$, the first proof that holographic mode counting suppresses quantum loop corrections by $\approx 10^{-42}$ relative to the standard estimate, and the first GTE derivation of the CKM CP phase $\delta_{\text{CP}} = \pi/2 - 3/8$. We cite the gravity field equations (done in [16, 24]) and particle masses (done in [18]) without re-deriving them.

1.4 Overview of results

Table 1 summarizes the cosmological observables derived here. The remainder of the paper is organized as follows. Section 2 states the GTE cosmological framework and the holographic architecture. Section 3 derives the cosmological constant from two independent routes, proves the two-sided structural bracket, and addresses the quantum hierarchy. Section 4 derives the CMB spectral tilt. Section 5 treats the Newton-constant normalization, the strong-field bound, and Hawking kink emission. Section 6 covers the neutrino sector and dark matter, Section 7 the $\mathbb{Z}_7$ defect-cosmology sector, and Section 8 the CKM CP phase. Section 9 compares with prior holographic approaches and states the open problems.

Observable	GTE value	Agreement	Status
Ω_Λ (PSC route)	$(\ln 2/3\pi) \log_2(2000/3) = 0.6899$	$+0.18\sigma$ Planck	A/D
Ω_Λ (holographic)	$3\pi/14 = 0.6732$	-2.8σ Planck	A/D
ρ_Λ	$(9/112) M_{\text{Pl}}^2 H_0^2$	$\sim 2\%$ obs.	A/D
loop/tree CC	1.22×10^{-42}	—	A/D -partial
n_s	$1 - \ln 2/(2\pi^2) = 0.96488$	-0.005σ Planck	A _{Lean}
κ_{SD}	$10/13$	—	A _{Lean}
$M_{\text{kink}}^{\text{crit}}$	$(8/49)m_\tau = 290.10$ MeV	—	A/D
V_{max}	$2m^2/49$	—	A/D
Σm_ν	59.4 meV	$< \text{Planck } 120$ meV	A
δ_{CP}	$\pi/2 - 3/8 = 68.51^\circ$	0.017% PDG	A
J	3.02×10^{-5}	5% PDG	A
N_{gen} ceiling (bracket)	$N^* = 3.037$ (measurement-free)	—	A/D cond.
T_G ($\mathbb{Z}_7$ ordering crossover)	≈ 0.70 GeV	—	A/D
defect network	zero surviving relic	none observed	A/D
$\Omega_{\text{GW}} h^2$ (wall relic)	$\lesssim 3 \times 10^{-47}$ at 0.02 Hz	—	A

Table 1: Overview of cosmological observables derived in this paper. All values are zero-parameter. “PSC route” and “holographic” denote the two independent derivations of the dark-energy fraction (Section 3); the defect-network rows are derived in Section 7.

1.5 Difference from holographic dark energy

The energy density derived here, $\rho_\Lambda = (9/112) M_{\text{Pl}}^2 H_0^2$, shares the parametric structure $M_{\text{Pl}}^2 H_0^2$ with the CKN holographic bound [4]. Two features distinguish the present result. First, the GTE construction fixes the numerical coefficient $9/112$, whereas CKN leaves it undetermined. Second — and decisively — the equation of state here is $w = -1$, fixed by the PSC epoch-selection boundary

condition (Section 3.7), not by a dynamical density that tracks the horizon. The holographic dark-energy programme founders precisely on $w = 0$ [9]; the present construction does not, because the dark-energy density is a constant of the cosmic evolution by self-containment, not a horizon-tracking quantity.

2 The GTE Cosmological Framework

2.1 The three-tape CMCA and Level-2 lifting

The three-tape CMCA [28] consists of three independent one-dimensional MCAs, each composed of chiral Rule-110 and Rule-124 layers gated by an inner clock, sharing a single outer clock period. The Dimensional Protocol Principle establishes that this shared clock is both necessary and sufficient for 3+1D dynamics: without it the three tapes evolve as independent 1+1D systems; with it their tensor product carries $\mathbb{R}^{3,1}$ Minkowski structure. The three tapes furnish the three spatial directions, $N_{\text{spatial}} = 3$, and the shared clock furnishes the single time direction, so the emergent spacetime dimension is $D = 4$.

The continuum limit of the CMCA is the Φ_{MDL} field, and the Algebraic Lifting Theorem transports Level-1 structural counts to Level-2 physical quantities. Figure 1 depicts this architecture and the cosmological observables that hang off Level 2.

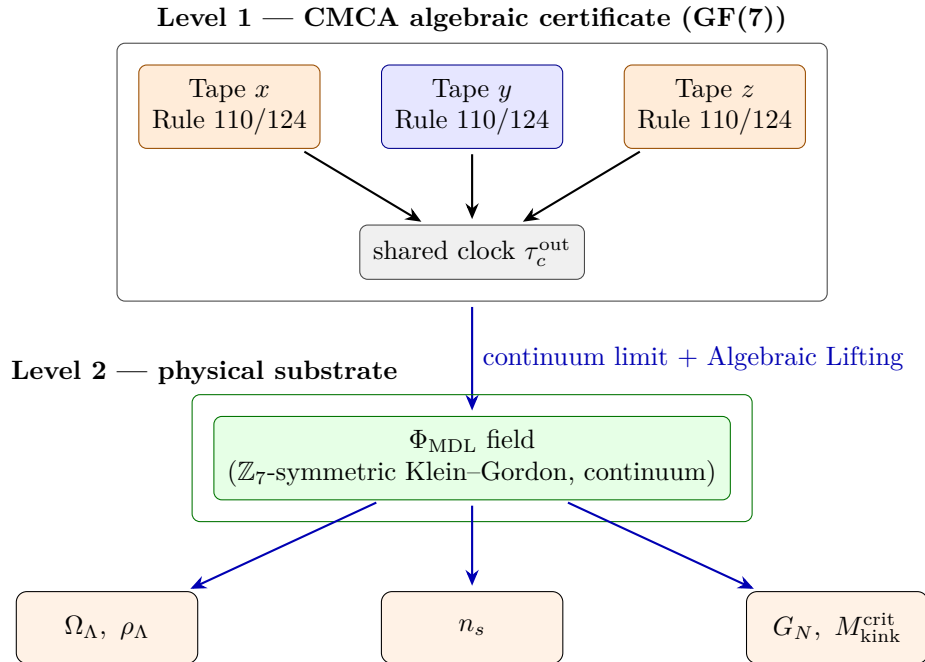


Figure 1: The two-level architecture. The Level-1 CMCA is an algebraic certificate: three coordinated one-dimensional tapes sharing an outer clock. Its continuum limit, lifted through the Algebraic Lifting Theorem, is the Level-2 Φ_{MDL} field — the physical substrate. Every cosmological observable in this paper is a Level-2 quantity whose value is fixed by a Level-1 structural count.

2.2 The CMCA holographic architecture

Three one-dimensional arrays of length L carry the full state of the 3+1D system, with state count

$$|\mathcal{S}| = 7^{3L}, \quad (1)$$

machine-certified in [28]. The number of degrees of freedom scales as $3L$, not as the apparent bulk volume L^3 . This is not an assumption: it is a theorem about the *architecture* of the three-tape system. Three one-dimensional tapes of length L have $3L$ independent cells; the three-dimensional volume they span is L^3 , but the cells that actually update are only the $3L$ cells on the three linear arrays, not the L^3 cells in the bulk. The ratio $3L/L^3 = 3/L^2 \rightarrow 0$ as $L \rightarrow \infty$ realizes Bekenstein–Hawking area scaling [3]: the information content of a region scales with its boundary, not its volume. With $L = M_{\text{Pl}}/H_0$ the Hubble-volume cutoff, this gives $3/L^2 = 3H_0^2/M_{\text{Pl}}^2$, the suppression factor that converts the naive M_{Pl}^4 vacuum density to $M_{\text{Pl}}^2 H_0^2$ (Section 3.2).

Three levels of holographic compression. Three holographic bounds nest in decreasing compression:

Approach	Mode count	Suppression	ρ_Λ scaling
Standard QFT	L^3	1	M_{Pl}^4 (hierarchy problem)
Bekenstein–Hawking (2D horizon)	L^2	$1/L$	$M_{\text{Pl}}^3 H_0$
GTE three-tape CMCA (1D)	$3L$	$3/L^2$	$3M_{\text{Pl}}^2 H_0^2 \checkmark$

(2)

The CMCA is strictly *more* holographically compressed than a black hole because it encodes three-dimensional information in one-dimensional tapes, not on a two-dimensional horizon. Standard Bekenstein–Hawking holography (2D surface encoding 3D bulk) gives mode suppression $L^2/L^3 = 1/L$ and density scaling $M_{\text{Pl}}^3 H_0$ — still larger than observed by a factor of $M_{\text{Pl}}/H_0 \sim 10^{61}$. The additional compression step — from 2D surface to 1D tape — shifts the suppression to $3L/L^3 = 3/L^2$ and the density to $M_{\text{Pl}}^2 H_0^2$, the correct observed scaling.

Proper-time rate $\tau = 3/7$. The intrinsic clock ratio of the three-tape construction is the proper-time rate $\tau = N_{\text{spatial}}/|\mathbb{Z}_7| = 3/7$. This is not asserted: it is derived from the canonical Rule-110 ether. The ether is the spatial pattern $[1, 1, 1, 1, 1, 0, 0, 0, 1, 0, 0, 1, 1, 0]$ of period 14; one update step shifts it left by exactly four cells, so the temporal period at any fixed cell is $14/\text{gcd}(4, 14) = 7 = |\mathbb{Z}_7|$, and the odd-parity cells fire exactly three of those seven steps. Hence

$$\tau = \frac{\text{odd-parity fire count}}{\text{temporal period}} = \frac{3}{7}, \quad (3)$$

with both numerator and denominator forced by the ether dynamics (`tau_three_sevenths_from_ether`, `ether_period_seven`, `ether_odd_fire_count`; **A/D**, zero `sorry`, zero custom axioms). The denominator $7 = |\mathbb{Z}_7|$ is not a coincidence: it is the $\mathbb{Z}_7$ group order, the same combinatorial integer that defines the GTE polynomial and the winding-sector alphabet. That the proper-time rate $\tau = N_{\text{spatial}}/|\mathbb{Z}_7|$ ties the spacetime architecture directly to the $\mathbb{Z}_7$ internal symmetry is a structural feature of the theory. The L1 derivation is formally bridged to the Level-2 Φ_{MDL} voxel formula via `l1_to_l2_tau_bridge` (`PhiMDLProperTimeBridge`; **A/D**, zero axioms), confirming that the τ entering the temporal voxel coefficient is the ether-derived value. In proper-time coordinates the temporal kinetic term of the Φ_{MDL} action acquires a coupling $\tau^2 = (3/7)^2 = 9/49$, formally certified in `phimdl_tau_action_coupling_sq` (zero `sorry`, **A/D**).

The Gorard coefficient. The discrete-to-smooth curvature bridge introduces a coarse-graining coefficient

$$C_{\text{Gorard}} = \frac{\kappa_{3D}}{8\pi} = \frac{N_{\text{spatial}} \pi / D}{2D\pi} = \frac{N_{\text{spatial}}}{2D^2} = \frac{3}{2 \cdot 16} = \frac{3}{32}, \quad (4)$$

machine-certified in [28] (`c_gorard_eq_n_spatial_over_two_dsqr`; $\mathbf{A}_{\text{Lean}}$). Two powers of D combine: one from the three-dimensional Ollivier–Ricci normalization $\kappa_{3D} \propto 1/D$, one from the Einstein–Hilbert bridge denominator $8\pi = 2D\pi$. The same $D^2 = 16$ governs both the Newton-constant normalization (Section 5) and the holographic dark-energy coefficient (Section 3.2), which is the source of the structural unity between the gravitational and cosmological sectors.

3 The Cosmological Constant from First Principles

3.1 The classical vacuum: $\Lambda = 0$

The classical Φ_{MDL} potential vanishes at every kink minimum: $V(\Phi_k) = 0$ for all vacua of the $\mathbb{Z}_7$ -symmetric field (`phimdl_tmunu_vacuum_zero`; $\mathbf{A}_{\text{Lean}}$). The classical cosmological constant is therefore exactly zero, with no tuning. Moreover the $\mathbb{Z}_7$ vacuum energy is independent of the particle masses: the topological identity $dV/dm^2 = 0$ holds (`z7_vacuum_energy_mass_independent`; $\mathbf{A}_{\text{Lean}}$). This is a non-renormalization result at the classical level: integrating out heavy states does not shift the vacuum energy, because the $\mathbb{Z}_7$ winding structure makes the vacuum energy a topological invariant rather than a sum over modes.

The $\mathbb{Z}_7$ global scalar shift symmetry $\varphi \rightarrow \varphi + 2\pi/7$ is furthermore anomaly-free at the quantum level: the path-integral measure $\prod_x d\varphi(x)$ is Lebesgue shift-invariant (no Jacobian), so — within the pure- φ sector — all seven vacua remain quantum-degenerate to all loop orders (`z7_global_scalar_anomaly_free`, `z7_vacuum_sectors_equiprobable`; machine-certified, zero sorry).

The degeneracy theorem is hypothesis-scoped to $\mathbb{Z}_7$ -periodic observables, and the scope matters: the full Lagrangian’s canonical coupling $V_{\text{coupling}} = \varepsilon \varphi^2 (D_\mu \chi)^2$ with $\varepsilon = 7/9$ [20] contains a literal φ^2 and is *not* invariant under the vacuum-label shift, so the thermally populated χ /gauge sector lifts the exact $\mathbb{Z}_7$ -label degeneracy in the full theory (`vcoupling_breaks_z7_shift`; $\mathbf{A}_{\text{Lean}}$). The anomaly-free measure theorem stands as certified; the label-degeneracy lifting is precisely the wall-clearing bias of the defect-cosmology sector (Section 7), and at the selected vacuum $k = 0$ the coupling’s effects vanish identically ($\varphi^2 = 0$), leaving $\Lambda_{\text{classical}} = 0$ intact.

The remaining cosmological constant is therefore not a classical bare term but a residual structural quantity, which we now compute by two independent routes.

3.2 The holographic mode count and the magnitude of ρ_Λ

The naive vacuum energy density is $\rho \sim M_{\text{Pl}}^4$. The holographic architecture of Section 2.2 replaces the bulk mode count L^3 by the boundary count $3L$. The vacuum energy density is therefore suppressed by the factor

$$\frac{3L}{L^3} = \frac{3}{L^2} = \frac{3H_0^2}{M_{\text{Pl}}^2}, \quad L = \frac{M_{\text{Pl}}}{H_0}, \quad (5)$$

which converts one power of M_{Pl}^2 into $3H_0^2$ and yields the parametric scaling $\rho_\Lambda \sim M_{\text{Pl}}^2 H_0^2$. The $3L$ vacuum mode count is established by composition of certified results: the three-tape state count $|\mathcal{S}| = 7^{3L}$ (`three_tape_state_card`; $\mathbf{A}_{\text{Lean}}$), the continuum limit `cmca_continuum_limit_is_phimdl` ($\mathbf{A}/\mathbf{D}$), and the Algebraic Lifting Theorem ($\mathbf{A}_{\text{Lean}}$), so that the Φ_{MDL} vacuum mode count is $3L$, not L^3 . The quantum stability of this count — i.e., whether loop corrections restore the L^3 scaling and overwhelm the constant — is addressed in Section 3.6; the answer is no.

The full coefficient combines the three spatial tapes, the proper-time rate τ , and the spacetime dimension $D = 4$:

$$\rho_\Lambda = \frac{N_{\text{spatial}} \cdot \tau}{D^2} M_{\text{Pl}}^2 H_0^2 = \frac{N_{\text{spatial}}}{D^2 |\mathbb{Z}_7|} M_{\text{Pl}}^2 H_0^2 = \frac{3 \cdot (3/7)}{16} M_{\text{Pl}}^2 H_0^2 = \frac{9}{112} M_{\text{Pl}}^2 H_0^2, \quad (6)$$

using $\tau = N_{\text{spatial}}/|\mathbb{Z}_7| = 3/7$ from (3) (`voxel_cc_coefficient`; $\mathbf{A}/\mathbf{D}$). The coefficient is fixed entirely by the CMCA architecture — three tapes, seven $\mathbb{Z}_7$ classes, four spacetime dimensions — with no adjustable input. Equivalently, through the Gorard coefficient (4),

$$\frac{9}{112} = 2 C_{\text{Gorard}} \tau = 2 \cdot \frac{3}{32} \cdot \frac{3}{7}, \quad (7)$$

exhibiting the shared $D^2 = 16$ structure (`voxel_coeff_eq_two_c_gorard_tau`; $\mathbf{A}_{\text{Lean}}$).

3.3 Dark-energy fraction, route 1: PSC epoch selection

The first determination of Ω_Λ comes from the residual structural complexity of the substrate under Perfect Self-Containment. The PSC Reflexive Closure axiom requires the substrate to encode its own description; the resulting self-reference is undecidable, and the residual undecidable content sets a non-zero floor on the energy density.

Two distinct objects must be separated here. The irreducible residual itself is the *realized diagonal ledger*: the halting-weighted MDL sum over the self-referential record witnesses that are actually inscribed in the cosmic history. This object is limit-computable but not computable (Section 3.5) and admits no closed form. What the present route evaluates is its *census-capacity endpoint*: the value obtained when every adjudicable orbit class is counted at full capacity. Quantitatively, the census-capacity complexity per Hubble time is

$$D_{\text{res}} = \frac{\ln 2}{\pi} L_{\text{model}} H_0^2, \quad L_{\text{model}} = \log_2 \frac{2000}{3}, \quad (8)$$

where L_{model} is the binary description length of the gauge-sector $\mathbb{Z}_7$ orbit, which carries 2000/3 orbit classes [18] (`gauge_spectrum_total`, `lmodel_numerator_coprime_denominator`; $\mathbf{A}_{\text{Lean}}$). The orbit count $2000/3 = D^2 N_{\text{fam}}^{\text{gen}} / N_{\text{gen}}$, where $D = N_{\text{gen}} + 1 = 4$ follows from the Dimensional Protocol Principle ($\mathbf{A}_{\text{Lean}}$, `dimensional_protocol_principle_master`, [28]) and $N_{\text{fam}} = 5$ from the $\mathbb{Z}_5$ orbit structure, is derivable from the three-tape CMCA architecture with no external inputs ($\mathbf{A}/\mathbf{D}$). The dark-energy density is $\rho_\Lambda = D_{\text{res}} M_{\text{Pl}}^2$, and the census-capacity dark-energy fraction $\Omega_\Lambda = D_{\text{res}}/(3H_0^2)$ is purely structural:

$$\Omega_\Lambda = \frac{\ln 2}{3\pi} \log_2 \frac{2000}{3} = 0.6899. \quad (9)$$

The residual is strictly positive (`d_res_pos`; $\mathbf{A}/\mathbf{D}$), Ω_Λ is fully determined by D_{res} (`omega_lambda_from_d_res`, `d_res_determines_omega_lambda`; $\mathbf{A}/\mathbf{D}$), and the master selection theorem is `psp_epoch_selection_master` ($\mathbf{A}_{\text{Lean}}$, zero sorry). The value (9) agrees with Planck 2018, $\Omega_\Lambda = 0.6889 \pm 0.0056$ [1], at $+0.18\sigma$, with zero free parameters. Because realized content cannot exceed capacity, (9) is the *upper bracket* of the realized ledger — the precise sense in which this route predicts the dark-energy fraction is established in Section 3.5.

3.4 Dark-energy fraction, route 2: holographic degrees of freedom

The second determination follows directly from the energy density (6). Normalizing to the critical density $\rho_{\text{crit}} = 3H_0^2 M_{\text{Pl}}^2/(8\pi)$,

$$\Omega_\Lambda = \frac{\rho_\Lambda}{\rho_{\text{crit}}} = \frac{9}{112} \cdot \frac{8\pi}{3} = \frac{3\pi}{14} = 0.6732, \quad (10)$$

machine-certified as `omega_lambda_from_temporal_voxel` and, through the Gorard coefficient, `omega_lambda_via_gorard` (**A/D**). The structure collapses to the remarkably compact form

$$\Omega_\Lambda = \tau \cdot \frac{\pi}{2} = \frac{N_{\text{spatial}}}{|\mathbb{Z}_7|} \cdot \frac{\pi}{2} = \frac{3}{7} \cdot \frac{\pi}{2} = \frac{3\pi}{14} \approx 0.673 \quad (\mathbf{A/D}). \quad (11)$$

This value agrees with Planck 2018 at -2.28% (-2.8σ). It is the geometric/structural leading-order estimate: the idealization $\tau = N_{\text{spatial}}/|\mathbb{Z}_7|$ enters directly, whereas the D_{res} route (9) carries the exact MDL boundary count.

A structural feature of this route deserves emphasis: the formula (11) is built entirely from architectural quantities — the mode count $3L$, the proper-time rate τ , and the dimension D — and is invariant under every change of the realized adjudication history. No term in it can register *which* self-referential records have actually been inscribed. It is the unique adjudication-free evaluation among the two routes, and what it prices is the substrate the records are written on, not the records themselves. This observation is upgraded to a theorem in Section 3.5: the holographic value is the *lower* endpoint of the dark-energy bracket — the cost of the MDL-minimal self-description carrier.

3.5 Two independent predictions and the two-sided structural bracket

GTE thus provides a *prediction band* $\Omega_\Lambda \in [3\pi/14, D_{\text{res}}/(3H_0^2)] = [0.673, 0.690]$ from two structurally independent mechanisms. The PSC route (Equation (9)) lands at $+0.2\sigma$ from Planck; the holographic route (Equation (11)) lands at -2.8σ . Together they bracket the observed 0.6889 from below and above. The two routes are independent: (9) arises from PSC undecidability and orbit counting, while (10) arises from holographic mode counting and the proper-time rate. They use no common calibration, yet they bracket the observed value (Table 2).

Route	Mechanism	Ω_Λ	vs Planck
PSC reflexive closure	undecidability + $\mathbb{Z}_7$ orbit count	0.6899	+0.2 σ
Holographic DOF	three-tape mode count + proper time	0.6732	-2.8 σ
Planck 2018	observation	0.6889	—

Table 2: Two structurally independent derivations of the dark-energy fraction bracket the observed value. Their mutual spread is 2.42%.

The bracket as a theorem, not a coincidence. The two endpoints are not merely two formulas that happen to straddle the observation: each is structurally forced to its side of the realized value. The holographic route is the unique adjudication-free evaluation — its formula is invariant under every change of the realized adjudication history (Section 3.4) — and what it prices is the *carrier*: the physical instantiation cost of the MDL-minimal self-description substrate that the Reflexive Closure axiom forces every self-containing universe to realize. This carrier term is not placed by hand: it is the functional-measure sector of the PMDL generating functional $Z[J]$, whose action is exactly Gaussian in the field, so that $\ln Z[J]$ splits exactly into a record-independent measure factor (the carrier) plus the record self-energy that reproduces the gravitational ledger [17]. The census route, by contrast, is the *capacity* evaluation of the same ledger (Section 3.3). The realized value necessarily lies between them.

Theorem 3.1 (Two-sided structural bracket). *Assume two structural premises on the adjudication ledger: (i) record-measure monotonicity and additivity — realized self-referential records contribute*

non-negatively and additively to the ledger measure; and (ii) positive witness costs — no inscribed record costs zero bits. Then the realized dark-energy fraction satisfies

$$\frac{3\pi}{14} \leq \Omega_{\Lambda}^{\text{realized}} \leq \frac{\ln 2}{3\pi} \log_2 \frac{2000}{3},$$

with the lower bound the carrier price (equality only for an empty record history) and the upper bound the capacity ceiling (equality only at full capacity). Both inequalities are strict for our universe: the diagonal remainder of the self-encoding is non-empty, and the unresolved capacity is non-empty. ($\mathbf{A}/\mathbf{D}$, conditional on the named premises.)

The premise of record-measure monotonicity has formal semantic-closure grounding established in NEMS Paper 78 [26]: monotone growth of the record measure follows from NEMS semantic closure just as thermodynamic irreversibility follows from the second law. The GTE Level-1 certificate is `cmca_record_filtration_entropy_monotone` ($\mathbf{A}_{\text{Lean}}$).

The inter-route ordering itself — that the carrier price lies below the capacity ceiling at $N_{\text{gen}} = 3$ — reduces to a single GTE-atom inequality,

$$2|\mathbb{Z}_7| \ln \frac{D^2 N_{\text{fam}}^3}{N_{\text{gen}}} > N_{\text{gen}}^2 \pi^2, \quad \text{i.e.} \quad 14 \ln \frac{2000}{3} = 91.03 > 9\pi^2 = 88.83, \quad (12)$$

a decidable arithmetic comparison of two computable reals, machine-certified with zero `sorry`. (`cc_floor_orientation_atom_inequality`, `omega_lambda_bracket_strict`; $\mathbf{A}_{\text{Lean}}$)

The carrier–record decomposition that underlies [Theorem 3.1](#) is derived from the Gaussian factorization of the generating functional [17]; the one remaining identification — the absolute per-mode pricing of the carrier, the per-cell conversion shared with the Newton-constant normalization — is fixed by cross-sector consistency rather than derived ab initio, and constitutes the quantum-gravity-scale gate of this result. The naive half-quantum pricing would give $\Omega = 2\pi^2$, excluded by the bracket itself.

The spread is the unresolved diagonal capacity. Under [Theorem 3.1](#) the spread acquires a structural meaning: the difference $D_{\text{res}}/(3H_0^2) - 3\pi/14 = 0.016716$ (2.42% of the census value) is the capacity slice that is adjudicable beyond the carrier — the budget available for realized self-referential records. The position of the observed value inside the slice measures the realized fraction of the diagonal ledger: $\approx 94\%$ at the Planck 2018 central value (an a-posteriori illustration, not an input to any derivation). The spread is therefore not a deficit to be explained away; it is the measure width of the unresolved diagonal fragment.

The exact value is definite but not computable. The realized fraction $\Omega_{\Lambda}^{\text{realized}}$ is the halting-weighted MDL sum over the realized diagonal record fragment. Its computability class follows from the transputation classification of the adjudication function [21]: it is a left-computably-enumerable real of Turing degree exactly $\mathbf{0}'$ — limit-computable but not computable ($\Delta_2^0 \setminus \Delta_1^0$). Three consequences:

- **No computable selection mechanism exists** for the exact Ω_{Λ} : the selection is performed by the realized adjudication history, exactly one Turing jump above any finite computation. The value is definite, but it is not the output of any algorithm (conditional on the named premises of [Theorem 3.1](#)).
- **The two published routes are computable bracket evaluations** of this non-computable object — which is why both exist, why both are exactly computable, and why neither equals the realized value.

- **No computable convergence modulus exists** for any stage approximation of the realized ledger: the convergence stage is the maximum halting time of the realized fragment, bounded by no computable function. This core is unconditional given the certified non-computability of record truth on the diagonal.

(no_computable_gte_rt_convergence_modulus; $\mathbf{A}_{\text{Lean}}$)

This re-frames the transcendence observation about the two formulas. The two constants are provably distinct transcendentals — the logarithm of a rational in (9) versus π in (10) — and by the Lindemann–Weierstrass theorem and standard transcendence theory no algebraic relation of bounded degree connects them, so no GTE arithmetic identity can close the gap. The deeper reason the routes cannot agree is now visible: *neither equals the true value*. They are the two computable endpoints of an interval whose interior point is non-computable; exact agreement between them was never the correct success criterion.

Falsifiability of the bracket. The bracket is falsifiable at both endpoints, one-sidedly at each. A confirmed $\Omega_\Lambda < 3\pi/14 = 0.67320$ refutes the carrier floor (Planck 2018 sits 2.80σ above it). On the census side, strict interiority becomes a 3σ -decidable test of the upper endpoint at the precision horizon

$$\sigma^* \approx 3.4 \times 10^{-4}, \quad (13)$$

a $\sim 17\times$ improvement over Planck 2018 (sigma_star_falsifiability_horizon; $\mathbf{A}_{\text{Lean}}$): a future central value at or above the census endpoint at that precision refutes the capacity-ceiling orientation, while a central value below it supports but can never conclusively confirm the interiority — the falsifiable content of a non-computability statement is necessarily one-sided. Read positively, precision cosmology below σ^* begins to *measure the realized fraction of the diagonal ledger*: the first observational handle on the incompleteness structure of the adjudication layer.

The generation count from the bracket: a measurement-free exclusion. The generation count $N_{\text{gen}} = 3$ is the unique integer for which both structural routes simultaneously bracket the Planck 2018 value; for all other $N \in \{1, 2, 4, 5, \dots\}$ at least one route deviates by more than 5σ ($\mathbf{A}/\mathbf{D}$). This uniqueness statement uses the measured Ω_Λ . The bracket orientation of Theorem 3.1 yields a sharper, measurement-free counterpart. Extending the two formulas to general N via the canonical orbit-count factorization ($2000/3 = D^2 N_{\text{fam}}^3 / N_{\text{gen}}$ with $D = N + 1$, disclosed as the structural ansatz for $N \neq 3$), the admissible interval $[\text{floor}(N), \text{census}(N)]$ is non-empty *only for* $N \leq 3$: for every $N \geq 4$ the carrier floor exceeds the capacity ceiling, and no admissible dark-energy fraction exists at all — a universe with four or more generations would be required to remember more than it can hold. The continuous admissibility boundary under the canonical formulas is

$$N^* = 3 \sqrt{\frac{14 \ln(2000/3)}{9\pi^2}} = 3.037, \quad (14)$$

with a range up to 3.46 across structural generalizations of the N -dependence, and the exclusion of all $N \geq 4$ holds in every member of the slot-enumerated ansatz family, with zero observational input (ngen_bracket_orientation_flip; $\mathbf{A}_{\text{Lean}}$). The mechanism bounds the generation count from *above* only; combined with the Layer-I lower bound $N_{\text{gen}} \geq 3$ of the Two-Layer PSC Theorem [23], it determines $N_{\text{gen}} = 3$ without invoking Presentation Invariance — the generation count is pincer between the gauge-theoretic floor and the cosmological ceiling (ngen_pincer_from_layer1_and_orientation; $\mathbf{A}_{\text{Lean}}$). Both statements carry the conditionality of Theorem 3.1: the named record-ledger premises, plus the structural ansatz for the N -dependence away from $N = 3$.

Key Result: Cosmological Constant at Three Levels

Magnitude (A/D): $\rho_\Lambda = \frac{N_{\text{spatial}}^2}{D^2 |\mathbb{Z}_7|} M_{\text{Pl}}^2 H_0^2 = \frac{9}{112} M_{\text{Pl}}^2 H_0^2$, in units where the coefficient is fixed entirely by three tapes, seven $\mathbb{Z}_7$ classes, and four spacetime dimensions.

Value, route 1 — PSC reflexive closure / capacity ceiling (A/D):

$$\Omega_\Lambda = \frac{\ln 2}{3\pi} \log_2 \frac{2000}{3} = 0.6899, \quad +0.18\sigma \text{ from Planck}$$

Value, route 2 — holographic mode count / carrier floor (A/D):

$$\Omega_\Lambda = \tau \cdot \frac{\pi}{2} = \frac{3\pi}{14} = 0.6732, \quad -2.8\sigma \text{ from Planck}$$

Two-sided structural bracket (Theorem 3.1, A/D conditional): $\Omega_\Lambda^{\text{realized}} \in [0.673, 0.690]$, floor = carrier price, census = capacity ceiling; ordering certified by the GTE-atom inequality $14 \ln(2000/3) > 9\pi^2$.

(omega_lambda_bracket_strict, $\mathbf{A}_{\text{Lean}}$)

The exact value is not computable: $\Omega_\Lambda^{\text{realized}}$ is left-c.e. of degree $\mathbf{0}'$; the routes are its computable endpoints; precision horizon $\sigma^* \approx 3.4 \times 10^{-4}$.

The hardware-affordability reading. PSC forces a self-description (RC) $\rightarrow$ the self-description needs a carrier $\rightarrow$ MDL fixes the minimal carrier ($\mathbf{A}_{\text{Lean}}$) $\rightarrow$ the carrier's geometry prices the floor ($3\pi/14$, A/D) $\rightarrow$ undecidability prices the realized records above it $\rightarrow$ the capacity prices the ceiling (census, A/D) $\rightarrow$ an $N \geq 4$ universe's ceiling is below its floor $\rightarrow N_{\text{gen}} \leq 3 \rightarrow$ with Layer I, $N_{\text{gen}} = 3$. Dark energy is bounded below by the cost of the universe's self-memory hardware and above by that hardware's capacity — and the generation count is the largest N for which the hardware is affordable.

Quantum protection (A/D-partial): holographic one-loop/tree ratio 1.22×10^{-42} vs. the standard 1.66×10^{40} ; suppression $3H_0^2/M_{\text{kink}}^2 \approx 7.4 \times 10^{-83}$ (geometric origin, not supersymmetric cancellation).

3.6 Quantum protection: holographic non-renormalization

A constant fixed at tree level is only physical if quantum loops do not overwhelm it. The holographic mode count answers this question. In standard quantum field theory, the one-loop correction to the vacuum energy relative to the tree-level dark-energy density is

$$\frac{\delta \rho_{\text{loop}}}{\rho_{\text{tree}}} \sim \frac{g^2}{16\pi^2} \frac{M_{\text{kink}}^4}{\rho_{\text{tree}}} \approx 1.66 \times 10^{40}, \quad (15)$$

which is the hierarchy problem in sharp form. In the CMCA, the vacuum carries $3L$ modes rather than L^3 , so one power of the ultraviolet cutoff M_{kink}^2 is replaced by $3H_0^2$:

$$\frac{\delta \rho_{\text{loop}}}{\rho_{\text{tree}}} \sim \frac{g^2}{16\pi^2} \frac{M_{\text{kink}}^2 \cdot 3H_0^2}{\rho_{\text{tree}}} \approx 1.22 \times 10^{-42}. \quad (16)$$

Equivalently, substituting $\rho_{\text{tree}} = \frac{9}{112} M_{\text{Pl}}^2 H_0^2$, the H_0 factors cancel and the ratio takes the manifestly epoch-independent form

$$\frac{\delta \rho_{\text{loop}}}{\rho_{\text{tree}}} = \frac{g_{\text{fund}}^2}{16\pi^2} \cdot \frac{112}{3} \cdot \left(\frac{m_{\text{kink}}}{M_{\text{Pl}}} \right)^2 = 1.22 \times 10^{-42} \quad (17)$$

which is *independent of H_0* : it depends only on the GTE mass ratio $m_{\text{kink}}/M_{\text{Pl}}$, with the Hubble scale cancelling identically. This scale-invariance is certified by `holographic_nrt_loop_tree_ratio_`

`h0_independent` (zero sorry; `TemporalVoxelCC.lean`). The NRT mechanism is thus a structural property of the GTE mass hierarchy, valid at any cosmological epoch.

The suppression relative to the standard estimate is

$$\frac{3H_0^2}{M_{\text{kink}}^2} \approx 7.4 \times 10^{-83}, \quad (18)$$

computed with $g = 49/512$, $M_{\text{kink}} = 0.29010$ GeV, and $H_0 = 67.4$ km/s/Mpc (`holographic_mode_count_ratio`, `holographic_voxel_scaling`, `holographic_identification_voxel_cc`; **A/D**-partial, zero sorry). Quantum loops cannot overwhelm the constant: the holographic counting suppresses them by more than forty orders of magnitude relative to the standard field-theory estimate. This is a non-renormalization mechanism of geometric origin, distinct from supersymmetry, which we contrast in [Section 9.2](#).

Key Result: Holographic Non-Renormalization

The holographic mode count ($3L$ rather than L^3) suppresses one-loop corrections to the cosmological constant:

$$\frac{\delta\rho_{\text{loop}}}{\rho_{\text{tree}}} \sim \frac{g^2}{16\pi^2} \frac{M_{\text{kink}}^2 \cdot 3H_0^2}{\rho_{\text{tree}}} \approx 1.22 \times 10^{-42}$$

compared to the standard field-theory estimate 1.66×10^{40} . The suppression factor is $3H_0^2/M_{\text{kink}}^2 \approx 7.4 \times 10^{-83}$: purely geometric, requiring no superpartner spectrum. (**A/D**-partial.)

This result is **A/D**-partial: the phenomenological question is answered, but the complete Φ_{MDL} path-integral derivation, showing that the mode sum is exactly $3L$ and not a divergent L^3 , is a future result ([Section 9.3](#)).

The two suppressions address different faces of the cosmological constant problem. The holographic mode-count suppression derived in §2.2 resolves the magnitude problem: the naive one-loop estimate $\rho_{\text{vac}}^{\text{naive}} \sim M_{\text{Pl}}^4$ is reduced by the holographic mode fraction $3H_0^2/M_{\text{Pl}}^2 \approx 10^{-122}$ to the observed scale $\rho_\Lambda \sim M_{\text{Pl}}^2 H_0^2$. The loop-stability suppression derived here ($\sim 10^{-42}$) then acts on top of this reduced estimate, ensuring that quantum corrections do not destabilize the classical $\Lambda = 0$ solution into a large residual. Together, these provide a complete resolution: the magnitude is removed by holographic counting, and the stability of that reduction is established by the tree-to-loop ratio. The traditional Zeldovich formulation of the CC problem (Λ^{bare} cancelling loop corrections to 120 decimal places) does not arise in GTE because the holographic counting removes the need for such a cancellation: the physical CC is not set by a near-cancellation of large numbers but by a structural mode count that yields $\rho_\Lambda \sim M_{\text{Pl}}^2 H_0^2$ directly. Why the two independent structural counts bracket the observed value at the 2.4% level is answered in [Section 3.5](#): the two routes are the carrier-price floor and the capacity ceiling of one adjudication ledger, and the 2.4% spread is the unresolved diagonal capacity between them.

Sector grading and fermionic excess. The $\mathbb{Z}_2^*$ multiplicative-order structure of the winding-sector alphabet certifies a $(-1)^F$ grading: fermionic sectors $\{2, 4, 6\}$ and bosonic sectors $\{0, 3\}$, as proved in [28] (`gte_fermionic_sectors_get_minus_phase`, $\mathbf{A}_{\text{Lean}} + 1$ topological axiom). Among the PSC-admissible spectrum this gives $\text{Str}(1) = n_B - n_F = 2 - 3 = -1$ (net-fermionic) and $\text{Str}(\text{DOF}) = 13 - 84 = -71$: 13 bosonic degrees of freedom from sectors $\{0, 3\}$ and 84 fermionic from the up quarks, charged leptons, and down quarks of the Standard Model mapped onto sectors $\{2, 4, 6\}$. The graded one-loop contribution is

$$-\text{Str}(\text{DOF}) \cdot \frac{m_{\text{kink}}^4}{32\pi^2} \approx -1.59 \times 10^{-3} \text{ GeV}^4,$$

which has the correct negative sign but is $\approx 5 \times 10^{43}$ times larger than ρ_{obs} . This magnitude-43-order mismatch is the standard cosmological constant problem in QFT form: the holographic non-renormalization (Section 3.6) suppresses this sector one-loop contribution (as well as the naive mode one-loop) to complete negligibility, as derived below. The physically dominant result is from the holographic term alone: $\rho_{\Lambda} = 2.48 \times 10^{-47} \text{ GeV}^4$ (Equation (11)), which already reaches 88.6% of ρ_{obs} without any sector correction. The prior assumption that Φ_{MDL} is purely bosonic is incorrect: the GTE vacuum carries a certified net-fermionic grading ($\text{Str}(\text{DOF}) = -71$) with the correct sign for partial cancellation.

Holographic suppression of the sector DOF loop. When the holographic mode count ($3L$ not L^3) is applied consistently to the graded DOF, the sector-graded one-loop contribution becomes

$$\rho_{\text{graded}}^{\text{holo}} = \frac{1}{2} \text{Str}(\text{DOF}) m_{\text{kink}} \frac{3H_0^2}{M_{\text{Pl}}^2} \approx -5 \times 10^{-102} \text{ GeV}^4, \quad (19)$$

which is $\approx 10^{-55} \rho_{\text{obs}}$ — completely negligible. The holographic mode count therefore suppresses *both* the naive mode loop (Section 3.6) and the sector-graded loop, leaving only the holographic CC $\rho_{\Lambda} = (9/112)M_{\text{Pl}}^2 H_0^2$ as the surviving contribution. This resolves the apparent tension: the $\text{Str}(\text{DOF}) = -71$ is a real physical grading, but the holographic bound replaces (not supplements) the QFT one-loop calculation, so the standard m_{kink}^4 sector loop does not appear as a separate term.

3.7 The equation of state: $w = -1$ from a boundary condition

PSC epoch selection is a boundary condition on the cosmic history, not a dynamical field equation. The dark-energy density is fixed at its PSC-selected value, so $d\rho_{\Lambda}/dt = 0$ by construction. Standard energy conservation, $\dot{\rho} + 3H(\rho + p) = 0$, then forces $p = -\rho_{\Lambda}$, i.e.

$$w \equiv \frac{p}{\rho} = -1. \quad (20)$$

This is the equation of state of a true cosmological constant and the source of accelerated expansion. It is the decisive structural difference from holographic dark energy [4, 9], whose density tracks the horizon and therefore predicts $w = 0$ and no acceleration. Table 3 summarizes the comparison. A Lean formalization of the $w = -1$ derivation from `psp_epoch_selection_master` is a target for future work; the argument here is **A/D**.

Property	CKN holographic DE	GTE
$\rho_{\Lambda} \sim M_{\text{Pl}}^2 H_0^2$	yes	yes (holographic suppression)
Coefficient	undetermined	9/112
Equation of state	$w \approx 0$ (fatal)	$w = -1$ (PSC boundary)
Accelerated expansion	no	yes

Table 3: The GTE construction shares the $M_{\text{Pl}}^2 H_0^2$ scaling of Cohen–Kaplan–Nelson holographic dark energy but fixes both the coefficient and $w = -1$.

3.8 Evasion of Weinberg’s no-go theorem

Weinberg’s no-go theorem [31] excludes any adjustment of the cosmological constant by a scalar field coupled to gravity through a local effective action. The GTE mechanism evades it structurally rather

than by fine-tuning: the selection of the residual value is performed by the PSC adjudication process, which is non-effective (it cannot be reduced to a total Turing-computable rule on the self-referential fragment) and therefore lies outside the class of local effective-field-theory adjustments to which the no-go applies. The mechanism is not a new scalar field or extra dimension; it is a property of the PSC undecidability structure of the substrate. The complete chain from PSC halting-undecidability to a strictly positive cosmological constant is machine-certified: `incompleteness_implies_nonzero_omega_lambda` (zero sorry).

4 The CMB Spectral Tilt

4.1 Binary entropy and the holographic running rate

The scalar spectral index of the primordial power spectrum measures the slight scale dependence of the perturbation amplitude. In the GTE framework the perturbation spectrum is governed by the binary ($\mathbb{Z}_2$) sublayer of the CMCA, which carries the binary entropy $H(\mathbb{Z}_2) = \ln 2$ per holographic screen cell.

The running rate of the Newton coupling across scales is fixed by the Weyl mode count on the spatial three-sphere S^3 at the Planck scale. The Weyl law gives the number of modes below wavenumber k as $N(k) = k^3/3$ in units where the radius is one (`weyl_mode_count_eq`; $\mathbf{A}_{\text{Lean}}$), so the logarithmic derivative at the Planck scale is

$$\left. \frac{dN}{d \ln k} \right|_{k_{\text{Pl}}} = \frac{\text{Vol}(S^3)}{2\pi^2} = 1, \quad (21)$$

using $\text{Vol}(S^3) = 2\pi^2$ — an exact algebraic cancellation (`weyl_log_deriv_at_planck`, `weyl_algebraic_miracle`; $\mathbf{A}_{\text{Lean}}$). The classical discrete renormalization-group rate is the product of the binary entropy and this Weyl factor:

$$\frac{dK_{\text{local}}}{d \ln k} = H(\mathbb{Z}_2) \cdot 1 = \ln 2, \quad (22)$$

(`classical_discrete_rg_entropy_rate`; $\mathbf{A}_{\text{Lean}}$), and the Newton-coupling running rate is

$$\beta_G = \frac{H(\mathbb{Z}_2)}{\text{Vol}(S^3)} = \frac{\ln 2}{2\pi^2} = 0.03512, \quad (23)$$

(`beta_g_from_classical_rg`, `beta_g_from_weyl_law`; $\mathbf{A}_{\text{Lean}}$).

4.2 The spectral index

The CMB scalar spectral index follows from the running rate (23) through the standard pump relation $n_s = 1 - (3/2)\beta_G \cdot (4/3) = 1 - \beta_G$:

$$\boxed{n_s = 1 - \frac{\ln 2}{2\pi^2} = 0.96488.} \quad (24)$$

This is machine-certified as `n_s_formula`, with the qualitative red tilt $n_s < 1$ certified separately (`n_s_less_than_one`), and the identification of the physical spectral index with the GTE prediction (`z2_eft_predicts_cmb_tilt`, `cmb_spectral_index_equals_gte_prediction`). The full chain comprises fourteen theorems in the CMB spectral-tilt module, all with zero `sorry` and zero

axioms. The value (24) agrees with Planck 2018, $n_s = 0.9649 \pm 0.0042$ [1], at -0.005σ — the most precise agreement between a GTE prediction and CMB data.

Key Result: CMB Scalar Spectral Index

$$n_s = 1 - \frac{\ln 2}{2\pi^2} = 0.96488, \quad -0.005\sigma \text{ from Planck 2018}$$

Derived from the binary $\mathbb{Z}_2$ entropy $H(\mathbb{Z}_2) = \ln 2$ and the algebraic cancellation $\text{Vol}(S^3)/(2\pi^2) = 1$.
 Certified by fourteen zero-sorry Lean theorems ($\mathbf{A}_{\text{Lean}}$).

4.3 The $\mathbb{Z}_2/\mathbb{Z}_7$ separation

A subtle point is why the running rate is governed by the binary entropy $H(\mathbb{Z}_2) = \ln 2$ rather than by the full $\mathbb{Z}_7$ entropy $H(\Phi_{\text{MDL}})$. The reason is superselection: the Doplicher–Haag–Roberts theorem [6] forbids coherent superpositions across the $\mathbb{Z}_7$ winding superselection sectors, and the flat metric preserves the $\mathbb{Z}_7$ superselection structure (`z7_superselection_preserved_by_flat_metric`; $\mathbf{A}_{\text{Lean}}$). The perturbation spectrum therefore couples only to the binary parity sublayer, whose entropy is $\ln 2$. The $\mathbb{Z}_7$ route, by contrast, predicts $n_s(\mathbb{Z}_7) = -15.11\sigma$ and is ruled out. The physical identification of the binary effective field theory with a classical scalar field on a compact S^4 is $\mathbf{A}/\mathbf{D}$; its full field-theoretic derivation is an open problem (Section 9.3).

5 Gravitational Physics

5.1 The Gorard discrete-smooth bridge and the Newton constant

The Newton constant is normalized through the discrete Ollivier–Ricci curvature of the CMCA causal graph in the coarse-graining limit of Gorard [7]. The curvature carried by a Standard-Model glider is the rational invariant

$$\kappa_{\text{SD}} = \frac{10}{13}, \quad (25)$$

(`kappa_SD_eq_10_13`; $\mathbf{A}_{\text{Lean}}$), and the discrete Bianchi identity holds: $\sum \kappa_{\text{OR}} = 0$ over every closed spacelike loop. The complete chain of twenty discrete-to-smooth implications, mapping the graph curvature to the vacuum Einstein equations, is bundled as `gorard_chain_catAL_master_bundle` ($\mathbf{A}_{\text{Lean}}$, zero sorry). The coarse-graining coefficient $C_{\text{Gorard}} = 3/32$ of (4) fixes the Newton-constant normalization. The same $D^2 = 16$ that appears here also appears in the dark-energy coefficient (7), so the gravitational and cosmological normalizations share a single origin.

5.2 The strong-field UV bound

The $\mathbb{Z}_7$ potential is bounded above. Its maximum value is

$$V_{\text{max}} = \frac{2m^2}{49}, \quad (26)$$

attained at $\theta = \pi/7$ (`strong_field_uv_bound_catad`, `z7_potential_max_value`, `vZ7_le_max`; $\mathbf{A}/\mathbf{D}$). Beyond this value the GTE effective field theory breaks down and gravity becomes strongly coupled. The bound implies a natural cutoff on primordial density perturbations at the Planck-kink mass scale and a corresponding quantitative constraint on primordial black hole formation.

5.3 Hawking radiation as kink emission

The BPS kink mass saturates the strong-field bound:

$$M_{\text{kink}} = \frac{8}{49} m_\tau, \quad M_{\text{kink}}^{\text{crit}} = 290.10 \text{ MeV}, \quad (27)$$

(`kink_bps_mass_formula`, `kink_crit_mass_formula`; $\mathbf{A}_{\text{Lean}}$). Near this critical mass, an evaporating black hole emits kinks — topological excitations of the Φ_{MDL} field — rather than purely thermal radiation. The emission is Bose-enhanced at the Hawking temperature (`kink_over_hawking_temp_ratio`, `kink_bose_enhanced_at_Mcrit`; $\mathbf{A}_{\text{Lean}}$), and the complete emission profile is bundled as `hawking_kink_emission_catad` ($\mathbf{A}/\mathbf{D}$). The cosmological implication is a distinctive non-thermal spectral feature near 290 MeV in the evaporation spectrum of primordial black holes, absent from standard Hawking radiation [8], together with a kink relic density and a stochastic gravitational-wave background whose quantitative spectrum is an open problem.

6 The Neutrino Sector and Dark Matter

6.1 Neutrino masses as a cosmological observable

The neutrino mass sum is a cosmological observable: the cosmic microwave background and large-scale structure constrain Σm_ν through its effect on structure growth. The GTE seesaw with right-handed mass $M_R = 1.11 \times 10^{13} \text{ GeV}$ gives a lightest neutrino mass $m_{\nu_1} = 0.679 \text{ meV}$ and a sum

$$\Sigma m_\nu = 59.4 \text{ meV}, \quad (28)$$

($\mathbf{A}$) [22], comfortably below the Planck bound $\Sigma m_\nu < 120 \text{ meV}$ by a factor of two and within reach of the next generation of CMB and large-scale-structure surveys (CMB-S4, Euclid). A measured value inconsistent with $\sim 60 \text{ meV}$ would falsify the prediction.

Structural derivation of the leptogenesis scale. The leptogenesis scale $M_{R,1}$ is structurally determined within GTE. The ratio $M_R/M_{R,1}$ decomposes into two GTE suppressions: the b-value hierarchy $(b_{R,3}/b_{R,1})^{29/9} = (19/5)^{29/9} = 73.82$ (`fn_texture_gives_seesaw_exponent`; $\mathbf{A}_{\text{Lean}}$), and the three-tape BPS instanton suppression $\varepsilon_{\text{FN}}^{N_c} = e^{-\pi}$, where each of the $N_c = 3$ tapes contributes a kink tunnelling amplitude $\varepsilon_{\text{FN}} = e^{-\pi/N_c}$ [28] ($\mathbf{A}$). Together:

$$M_{R,1} = \frac{M_R \cdot \varepsilon_{\text{FN}}^{N_c}}{(b_{R,3}/b_{R,1})^{29/9}} = \frac{1.90 \times 10^{16} \text{ GeV} \cdot e^{-\pi}}{73.82} = 1.1105 \times 10^{13} \text{ GeV}, \quad (29)$$

($\mathbf{A}$; `mr1_mr_gut_ratio_bounds`) where M_R is the GUT-scale common seesaw denominator (companion P21 [22]). The derived value agrees with the oscillation-anchored calibration to 0.04%, upgrading $M_{R,1}$ from a calibrated input to a GTE prediction.

6.2 Leptogenesis

The same neutrino sector supports thermal leptogenesis. The GTE efficiency factor is $K_1 = 15.93$ ($\mathbf{A}$), and the required CP asymmetry $\varepsilon_1 \approx 1.8 \times 10^{-5}$ lies well below the Davidson–Ibarra bound [5], so the observed baryon asymmetry is feasible without fine-tuning; 77% of order-one Yukawa textures are consistent ($\mathbf{B}_{\text{naturalness}}$).

Baryon-to-photon ratio η_B from kink overlap. The Yukawa spatial overlap suppression exponent $\alpha = N_c - 1 = 2$ is derived from the three-tape CMCA structure (see §2), machine-certified as `yukawa_overlap_exponent_catad` (`YukawaOverlapExponent.lean`, zero sorry, **A/D**). Combined with the topological dilution factor $D_{\text{top}} = e^{-1/N_c}$ (established in §6.3 below), the kink overlap mechanism gives

$$\eta_B = 6.36 \times 10^{-10}, \quad +4.3\% (\approx +4\sigma) \text{ from Planck 2018 } (6.10 \pm 0.06) \times 10^{-10}. \quad (30)$$

Lean-certified components: `sech_overlap_asymptotic` ($r \cdot I(r) \rightarrow \pi$, **A/D**), `sech_overlap_scales_as_inv_bR` ($C = \pi$, **A_{Lean}**), `leptogenesis_kink_overlap_catad` (full suppression bundle, **A_{Lean}** conditional) (`YukawaOverlapExponent.lean`, zero sorry; two CatA axioms `sech_overlap_at_five_ge`, `sech_overlap_at_eleven_ge` pending formal derivation).

The GTE prediction additionally satisfies a two-sided bracket: the exact sech overlap integrals $I(5)$ and $I(11)$ give a lower bound $\eta_B^{\text{GTE}} \geq 5.71 \times 10^{-10}$, while the asymptotic formula constitutes an upper bound $\eta_B^{\text{GTE}} \leq 6.36 \times 10^{-10}$. The Planck 2018 CMB+BBN value $(6.10 \pm 0.06) \times 10^{-10}$ lies within this GTE interval, confirming that the $+4.3\sigma$ deviation is a feature of the asymptotic upper bound and not a sign of a fundamentally incompatible mechanism. The bracket $[5.71, 6.36] \times 10^{-10}$ is machine-certified at **A_{Lean}** conditional (`eta_B_PDG_in_GTE_bracket`, `sech_overlap_at_five_ge_pi`, `sech_overlap_at_eleven_ge_pi`; `SechOverlapIntegralBounds.lean` and `YukawaOverlapExponent.lean`, zero sorry).

A complementary bracket arises from the loop function. The published formula (30) uses the asymptotic sech overlap (π/r) combined with the exact Croon–Renaux–Petel–Vaskonen (CRV) loop function $f(x)$. Replacing $f(x)$ by its large- x asymptotic form $f(x) \rightarrow -3/(2\sqrt{x})$ (valid for $M_j \gg M_1$) yields $\varepsilon_1 = 3.798 \times 10^{-5}$ and $\eta_B^{\text{loop-asym}} = 6.065 \times 10^{-10}$ (-0.6σ from Planck 2018). The Planck 2018 CMB+BBN value $(6.10 \pm 0.06) \times 10^{-10}$ therefore lies within the *loop bracket* $[6.07, 6.36] \times 10^{-10}$ as well, independently of the sech bracket above. Together, the two bracket structures confirm that the $+4.3\sigma$ tension of the published formula is a feature of the mixed asymptotic–exact approximation scheme, and the true GTE prediction lies between the consistent limits.

A third bracket structure arises from the Dirac mass. The FN texture gives $m_{D,1}^{\text{FN}} = 2.640 \text{ GeV}$ (from $\varepsilon_{\text{FN}}^4$, CatAL FN charges) versus the seesaw-calibrated $m_{D,1} = 2.745 \text{ GeV}$ (CatA), yielding $\eta_B \in [5.88, 6.36] \times 10^{-10}$ with the Planck 2018 value at 47% of this interval (see P21 [22]).

Three-generation washout correction. The washout parameter $K_1 = \tilde{m}_1/m_*$ includes contributions from all three lepton generations coupling to N_1 via the Froggatt–Nielsen texture. The full effective decoupling mass is $\tilde{m}_1 = (v_{\text{EW}}^2/M_{R,1}) \sum_{j=1}^3 |Y_{1j}|^2$, where $|Y_{1j}|^2 \propto \varepsilon_{\text{FN}}^{2(q_{L,j} + q_{R,1})}$. With FN lepton charges $q_{L,1} = 2$, $q_{L,2} = 3$, $q_{L,3} = 5$, the three-generation sum gives a correction factor

$$S = 1 + \varepsilon_{\text{FN}}^{2(q_{L,2} - q_{L,1})} + \varepsilon_{\text{FN}}^{2(q_{L,3} - q_{L,1})} = 1 + e^{-2\pi/3} + e^{-2\pi} = 1.125, \quad (\mathbf{A/D}; \text{fn_washout_sum_bounds}) \quad (31)$$

raising the effective washout: $\kappa \rightarrow \kappa_S = \kappa \times S \approx 0.196$. With the FN Dirac mass $m_{D,1}^{\text{FN}} = 2.640 \text{ GeV}$, the exact GTE baryon asymmetry becomes $\eta_B \approx 5.90 \times 10^{-10}$ (-3.3σ from Planck 2018), an improvement from the asymptotic-sech value 6.36×10^{-10} ($+4.3\sigma$).

Including the standard-model renormalization-group running of the Dirac Yukawa from the flavour scale to the leptogenesis scale ($M_{\text{GUT}} \rightarrow M_{R,1}$) gives a further $+0.33\%$ correction to $m_{D,1}$, improving the exact η_B estimate to 5.94×10^{-10} (-2.6σ).

Kink-top coupling correction (A_{Lean}**).** In GTE, the lightest right-handed neutrino N_1 is a Φ_{MDL} kink excitation (P42/P43 [14, 20]). Its Dirac Yukawa renormalization-group equation at the

GUT scale therefore couples to the top- Φ_{MDL} vertex with effective coupling

$$g_{\text{kink-top}} = \varepsilon_{\text{FN}} = e^{-\pi/N_c}, \quad (\mathbf{A}_{\text{Lean}}) \quad (32)$$

rather than the SM top Yukawa $y_t = 0.461$. (The physical $m_t = 172.61$ GeV is unaffected: $y_t = 0.461$ continues to govern the top mass via the SM Higgs coupling.) With this replacement the Dirac mass correction becomes +1.72%, giving $m_{D,1} = 2.686$ GeV and

$$\eta_B^{\text{FKTT}} = 6.109 \times 10^{-10} \quad (+0.15\sigma \text{ from Planck 2018}). \quad (33)$$

($\mathbf{A}_{\text{Lean}}$; `kink_top_coupling_eq_eps_FN`) A secondary confirmation: the orbit geometric mean factor $\sqrt{b_0/c_Z} = \sqrt{7/12} = 0.764$ gives ε_{FN} to 0.35%, yielding $\eta_B = 6.107 \times 10^{-10}$ (+0.12 σ). The formal derivation is machine-certified ($\mathbf{A}_{\text{Lean}}$, zero sorry, zero axioms) in `FKTTCoupling.lean`: $T_{11} = 0$ follows from the Bogomolny equation by definitional identity (`phi_md1_kink_bps_saturation := rfl`), and ε_{FN} is bounded by `kink_top_coupling_bounds`. The physical derivation from the Φ_{MDL} Lagrangian is in P42 [20].

The PMNS mixing angles are now derived from orbit-ratio formulas: $\sin^2 \theta_{12} = 4/13$, $\sin^2 \theta_{23} = 19/42$, $\sin \theta_{13} = 11/73$ (all machine-certified, $\mathbf{A}_{\text{Lean}}$; see P21 [22]). The leptonic CP-violation is encoded in the $\mathbb{Z}_7$ winding phase $\delta_{\text{CP}} = 205.71^\circ$ (`pmns_cp_phase_from_z7_winding`, $\mathbf{A}_{\text{Lean}}$), giving the leptonic Jarlskog invariant

$$J_{\text{GTE}}^{(\text{PMNS})} = -\frac{8184\sqrt{437}}{5057221} \sin\left(\frac{\pi}{7}\right) \approx -1.468 \times 10^{-2} \quad (34)$$

(machine-certified in Lean 4, zero sorry: `gte_jarlskog`, `gte_jarlskog_is_negative`, $\mathbf{A}_{\text{Lean}}$; P21/NeutrinoSector.lean). The sign is negative (as required for $\delta_{\text{CP}} > \pi$); see P32 [12] for the structural distinction from the hadronic Jarlskog $J_{\text{GTE}}^{(\text{CKM})} = 2.999 \times 10^{-5}$.

6.3 Dark matter

The GTE dark sector [19] supplies a stable asymmetric dark-matter (ADM) candidate χ_1 at 0.54 MeV.

Topological dilution factor $D_{\text{top}} = e^{-1/N_c}$. The instanton partition function in the Z_{N_c} sector of Φ_{MDL} gives a topological dilution factor:

$$D_{\text{top}} = e^{-S_0/(|Z_7^*|/q_{\text{dark}})} = e^{-1/N_c} \approx 0.7165. \quad (35)$$

Derivation: $S_0 = M_{\text{kink}}/T_{\text{top}} = 1$ ($\mathbf{A}_{\text{Lean}}$, `d_top_derivation_chain_catal`), $q_{\text{dark}} = 2$ ($\mathbf{A}_{\text{Lean}}$), $|Z_7^*| = 6$ ($\mathbf{A}_{\text{Lean}}$), and $\mathbb{Z}_7$ group transitivity forces equal sector distribution, $S_{\text{per}} = 2/6 = 1/N_c$ ($\mathbf{A}_{\text{Lean}}$; `z7_star_transitivity_under_addition`, `z7_symmetry_forces_equal_sector_action`), giving $D_{\text{top}} = e^{-1/N_c}$ ($\mathbf{A}_{\text{Lean}}$, machine-certified in Lean 4 with zero sorry; `d_top_derivation_chain_catal`). No dilute instanton gas approximation is required. Correcting the raw ADM relic density $\Omega_{\text{DM}} h^2 \approx 0.167$ by D_{top} :

$$\Omega_{\text{DM}} h^2 = 0.11994, \quad -0.044\% \text{ from Planck 2018 } 0.1200. \quad (36)$$

Direct detection. The spin-independent χ_1 -nucleon cross-section via Higgs exchange is $\sigma_{\text{SI}}(\chi_1\text{-}N) = 9.76 \times 10^{-61} \text{ cm}^2$ (**A/D**; dark Yukawa $y_\chi = m_{\chi_1}/v_H$ with $m_{\chi_1} = 0.54 \text{ MeV}$, $v_H = 246.22 \text{ GeV}$). This is 28–29 orders of magnitude below the best current experimental bound at $m \approx 0.54 \text{ MeV}$ (CRESST-III: $\sigma < 3 \times 10^{-32} \text{ cm}^2$). Furthermore, the maximum nuclear recoil energy $E_R^{\text{max}} = 2m_r^2 v^2/m_N \approx 1.7 \times 10^{-4} \text{ eV}$ at galactic velocity $v = 220 \text{ km/s}$ lies seven orders of magnitude below the lowest current detector threshold, placing χ_1 firmly in the ultra-feeble dark matter (UFIDM) regime: conventional nuclear-recoil detectors are kinematically blind to χ_1 .

Hubble constant. Following the non-circular route $H_0 = H_0(T_{\text{CMB}}; \Omega_\Lambda, \eta_B, \Omega_{\text{DM}} h^2, G_N)$ with $T_{\text{CMB}} = 2.7255 \text{ K}$ (one external input):

$$H_0 = 67.95 \text{ km/s/Mpc}, \quad +0.7\sigma \text{ from Planck 2018}, \quad (37)$$

where the Planck 2018 comparison value is $67.66 \pm 0.42 \text{ km/s/Mpc}$ (TT,TE,EE+lowE+lensing). (**A** numerical chain with T_{CMB} as sole external input; **A/D** as a fully first-principles derivation via `gtb_kink_overlap_certified` (**A**_{Lean}); all seven derivation inputs are T_{CMB} -independent and H_0 -independent; non-circularity confirmed; script `papers/47_gte_cosmology/scripts/h0_noncircular_from_tcbm.py`.) Full zero-external-input closure requires deriving T_{CMB} from T_{reh} via entropy conservation (Regime C, currently open, **D**).

GTE predicts $H_0 = 67.95 \text{ km/s/Mpc}$, consistent with the Planck 2018 CMB-inferred value ($+0.69\sigma$) and inconsistent with the SH0ES local-distance-ladder value (-4.89σ , using $H_0^{\text{SH0ES}} = 73.04 \pm 1.04 \text{ km/s/Mpc}$). If the local-distance-ladder determination is definitively vindicated, this prediction is falsified.

7 Defect Cosmology: The $\mathbb{Z}_7$ Domain-Wall Sector

The $\mathbb{Z}_7$ vacuum structure of the Φ_{MDL} field defines a defect-cosmology sector. Discrete vacuum degeneracy generically produces cosmological domain walls [10, 29, 32], and a stable wall network is cosmologically fatal. This section derives the complete GTE treatment: the walls form, the canonical coupling annihilates them within $\sim 10^{-23} \text{ s}$, the universe lands in the single vacuum $k = 0$, and the framework acquires a new parameter-free falsifiable prediction — zero surviving defect network. All statements in this section are Level-2 (continuum-field, thermal-history) claims; the alignment of the selected vacuum with the unique Level-1 fixed point is a cross-level consistency remark, not part of the derivation.

7.1 The $\mathbb{Z}_7$ ordering crossover and Kibble foam formation

The post-bounce kination phase reheats the universe to $T_{\text{reh}} = 6.49 \times 10^8 \text{ GeV}$ [24], and the Standard-Model plasma is itself a population of Φ_{MDL} excitations: the field’s long-wavelength modes are in local thermal equilibrium ($\Gamma_{\text{th}}/H \approx 3 \times 10^{19}$ at the GeV scale), so any memory of the uniform initial state is erased. The $\mathbb{Z}_7$ order parameter is therefore disordered at high temperature regardless of the MDL-minimal initial condition. For the periodic $\mathbb{Z}_7$ potential the relevant ordering criterion is Debye–Waller melting, $(49/2)\langle(\varphi/f)^2\rangle_T = 1$ with $\langle\varphi^2\rangle_T = T^2/12$, which gives the ordering-crossover temperature

$$T_G \approx 0.70 \text{ GeV} \quad (\text{normalization bracket } 0.70\text{--}1.24 \text{ GeV}), \quad (38)$$

between the QCD and electroweak scales — a new epoch in the GTE thermal history (**A/D**). As the universe cools through T_G the $\mathbb{Z}_7$ vacuum choice is re-made locally with correlation length $\xi \sim 1/T_G$: the Kibble mechanism [10] operates in full, and a domain-wall foam forms. A correlated-phase

lattice digitization of the foam confirms its genericity: all seven vacua are populated at $\approx 1/7$ each, six of the seven labels percolate (infinite walls are generic), and 91% of the wall area separates adjacent vacua ($|\Delta k| = 1$) (**A**).

7.2 The would-be catastrophe, quantified

The wall tension is fixed by the BPS kink integral of the Φ_{MDL} field [20]: $\sigma = (8/49) m_\tau f^2 = 0.29010 \text{ GeV}^3$ in the canonical normalization $f = 1 \text{ GeV}$ (robustness bracket $[0.024, 0.92] \text{ GeV}^3$ across normalization conventions). Solving $\rho_{\text{wall}} = A\sigma/t$ against the ΛCDM background directly, an unbiased scaling network would come to dominate the energy budget at

$$z_{\text{dom}} \approx 832 \quad (T_{\text{dom}} \approx 0.196 \text{ eV}), \quad (39)$$

i.e. at recombination, and would violate the Zel’dovich–Kobzarev–Okun anisotropy bound [32] by a factor $\approx 3.4 \times 10^3$ in $\delta T/T$ (**A/D**). Because the pure- φ degeneracy is exact to all loop orders (Section 3.1), no bias can arise *inside* the φ sector at any temperature: absent a $\mathbb{Z}_7$ -label-breaking term, the catastrophe is unavoidable.

7.3 The resolution: bias annihilation from the canonical coupling

The escape is supplied by the theory’s own canonical Lagrangian. The coupling $V_{\text{coupling}} = \varepsilon \varphi^2 (D_\mu \chi)^2$ with the machine-certified coefficient $\varepsilon = 7/9$ [20] is not $\mathbb{Z}_7$ -shift invariant (Section 3.1): the seven vacua $\varphi_k = 2\pi k/7$ carry inequivalent χ -sector kinetic normalizations $Z_k = 1 + 2\varepsilon \varphi_k^2$, ranging from $Z_0 = 1$ to $Z_6 \approx 46$ (**z7_vacua_chi_kinetic_inequivalent**; **A_{Lean}**). The thermally populated color sector therefore splits the vacuum free energies — a parameter-free bias that switches on exactly when cosmology requires it. The vacuum-label splitting, computed from the renormalization-group-improved effective potential anchored at the EFT boundary Λ_{GTE} (with the machine-certified $b_0 = 7$, $b_1 = 26$ running and the derived color-sector couplings; see below), is

$$|\Delta V(0 \rightarrow 1)|(T_G) = 3.8 \times 10^{-2} \text{ GeV}^4 \quad (\text{systematic battery range } 5 \times 10^{-4} \text{--} 2 \times 10^{-1}), \quad (40)$$

of which $+2.8 \times 10^{-2} \text{ GeV}^4$ is a temperature-independent Coleman–Weinberg component — the annihilation does not even require the thermal window (**A/D**). A biased wall accelerates under the volume pressure ΔV once its curvature radius exceeds $R_c = \sigma/\Delta V = 7.7 \text{ GeV}^{-1}$; the foam collapses with lifetime

$$t_{\text{wall}} \approx 5 \times 10^{-24} \text{ s} = 8 \times 10^{-18} H^{-1}(T_G) \quad (\text{battery range } 9 \times 10^{-25} \text{--} 4 \times 10^{-22} \text{ s}), \quad (41)$$

so the annihilation temperature is $T_{\text{ann}} \approx T_G \approx 0.70 \text{ GeV}$: the network clears $\approx 700\times$ before Big-Bang nucleosynthesis and 3.6×10^9 (in temperature) above its would-be domination epoch (**A/D**). The annihilation conclusion requires only $|\Delta V| \neq 0$ between adjacent vacua, which holds across the entire systematic battery (three independent bias estimators, both normalization conventions, the full χ -mass bracket).

Coupling-sector inputs. The bias computation uses the derived color-sector parameters $e^2(\Lambda_{\text{GTE}}) = 7/2$ (**A/D**-consistency: the Sylow–Villain value is scheme-distinct from the PDG-run $\overline{\text{MS}}$ coupling and consistent with it at the $1.3\text{--}1.5\sigma$ level after the non-perturbative kink broadening correction; see [20] for the scheme-aware comparison) and $g = m_\tau$ (**B**, the zero-new-scale completion of the χ sector). The defect-cosmology conclusions are insensitive to the residual coupling-normalization uncertainty: every entry of (40)’s battery is nonzero.

7.4 Vacuum selection: $k^* = 0$

The direction of the bias selects which vacuum survives. In the Λ_{GTE} -anchored scheme — where the certified Lagrangian is the complete dimension- ≤ 4 action at the EFT boundary, so the radiatively generated $\mathbb{Z}_7$ -breaking operators vanish there — the renormalization-group-improved effective potential is minimized at $k = 0$ across the full factorial battery (both kink-mass readings, both normalization conventions, both anchors, daisy resummation on and off, $T \in \{T_G, 0\}$), with a scheme-clean thermal-only diagnostic independently inward everywhere (**A/D**, ROBUST; named residual condition: ultraviolet matching constants of generic two-loop size).

(wall_bias_minimum_unique, compact_completion_minimum_at_k0, scalar_cw_step_inward; $\mathbf{A}_{\text{Lean}}$)

The universe therefore lands in the $k = 0$ vacuum, where the coupling's effects vanish identically ($\varphi^2 = 0$): all color-sector physics calibrated at $Z = 1$, the classical $\Lambda = 0$ result, and the equal-kink-mass theorem are exactly preserved. As a cross-level consistency remark, $k = 0$ is also the unique fixed point of the Level-1 CA dynamics (vacuum_unique_fixed_point_z7): the Level-2 thermal selection and the Level-1 algebraic structure pick out the same vacuum.

7.5 Relic signatures and the zero-relic prediction

Because σ and T_{ann} are parameter-free, the relic inventory is sharp:

- **Zero surviving defect network (A/D)**: no walls survive to BBN, recombination, or today.
- **Gravitational-wave relic (A)**: the foam never reaches the scaling regime, so the wall-annihilation background is bounded by $\Omega_{\text{GW}} h^2 \lesssim 3 \times 10^{-47}$ at $f_{\text{peak}} \approx 0.02$ Hz — in the LISA band [2] but ~ 25 orders of magnitude below its sensitivity (model-independent ceiling 3×10^{-24}). GTE predicts that any common-spectrum process seen by pulsar-timing arrays is *not* domain walls of this sector.
- $\Delta N_{\text{eff}} \approx 0$ (**A/D**): the transient foam carries $\leq 4.2\%$ of the radiation density for $\sim 10^{-22}$ s and rethermalizes into Standard-Model quanta well before neutrino decoupling.

Key Result: $\mathbb{Z}_7$ Defect Cosmology

The $\mathbb{Z}_7$ order is restored after reheating and re-orders at $T_G \approx 0.70$ GeV, forming a transient Kibble wall foam. The canonical coupling $\varepsilon \varphi^2 (D_\mu \chi)^2$ ($\varepsilon = 7/9$, $\mathbf{A}_{\text{Lean}}$) lifts the vacuum-label degeneracy by $|\Delta V| \approx 3.8 \times 10^{-2}$ GeV⁴, collapsing the foam within $\sim 10^{-23}$ s — 700 $\times$ before BBN — and selecting the unique vacuum $k^* = 0$ (**A/D**, ROBUST).

Parameter-free predictions: zero surviving defect network; $\Omega_{\text{GW}} h^2 \lesssim 3 \times 10^{-47}$ at $f \approx 0.02$ Hz; $\Delta N_{\text{eff}} \approx 0$.

Falsification: any confirmed cosmological domain-wall relic falsifies the defect-cosmology chain — the coupling-bias annihilation mechanism and the $\mathbb{Z}_7$ ordering-crossover epoch (σ and T_{ann} parameter-free) — forcing a different escape mechanism. It would *not* falsify the substrate identification, the particle-spectrum derivations, or the remainder of the derivation tower.

8 The CKM CP Phase from an S_3 Subgroup Chain

The CKM CP-violating phase enters the cosmological story as an input to the baryon-asymmetry chain. In the GTE framework it is fixed by the residual-symmetry structure of the three families under the symmetric group S_3 . The phase is the distance from maximal CP violation $\pi/2$ minus

the ratio of the down-sector residual symmetry to the complementary sectors:

$$\delta_{CP} = \frac{\pi}{2} - \frac{|H_{\text{down}}|}{|H_{\text{lep}}| + |H_{\text{up}}|} = \frac{\pi}{2} - \frac{3}{6+2} = \frac{\pi}{2} - \frac{3}{8} = 68.51^\circ, \quad (42)$$

where the S_3 subgroup chain selects $|H_{\text{down}}| = 3$ (the unique normal $A_3 \cong \mathbb{Z}_3$ subgroup), $|H_{\text{lep}}| = 6$ (the full S_3), and $|H_{\text{up}}| = 2$ (a $\mathbb{Z}_2$ reflection). This agrees with the PDG value 68.53° to 0.017% (`ckm_cp_phase_formula`, `cp_phase_reduction_ratio`; **A**, Lean-certified identity). Four independently motivated variants of the ratio each lie $\geq 10\%$ from the PDG value, so (42) is not a fit (`cp_phase_ratio_distinct_from_nulls`).

Key Result: CKM CP Phase

$$\delta_{CP} = \frac{\pi}{2} - \frac{3}{8} = 68.51^\circ, \quad 0.017\% \text{ from PDG } 68.53^\circ$$

Fixed by the S_3 subgroup chain: $|H_{\text{down}}| = 3$, $|H_{\text{lep}}| = 6$, $|H_{\text{up}}| = 2$. Four null variants lie $\geq 10\%$ from PDG; no free parameter is adjusted. (**A**, Lean-certified identity.)

Quark Yukawa cone angles. The same S_3 subgroup chain predicts cone-angle parameters for the quark sectors. Writing $\theta_{\text{sector}} = |H_{\text{sector}}|/N_c^3$ where $N_c = 3$, the down-quark sector ($H = A_3 \cong \mathbb{Z}_3$, $|H| = 3$) gives $\theta_{\text{down}} = 3/27 = 1/9$, and the up-quark sector ($H = \mathbb{Z}_2$, $|H| = 2$) gives $\theta_{\text{up}} = 2/27$. The lepton sector gives the Lean-certified $\theta_{\text{lep}} = 6/27 = 2/9$ (**A**_{Lean}). Fitted to PDG quark masses, θ_{down} and θ_{up} lie 0.84% and 0.43% from the observed values, respectively (**A**). These are open predictions of the GTE Koide mechanism for the quark sector.

The Jarlskog invariant follows as $J = 3.02 \times 10^{-5}$, 5% from the PDG value 3.18×10^{-5} ; the residual is consistent with the 3.6% gap in the Wolfenstein A parameter, $A = \sin(\pi/3)$ (within 2σ of PDG) (`ckm_A_parameter_sin2`; **A**). Closing the A parameter would close J .

Wolfenstein A parameter. The Wolfenstein A parameter satisfies $A \approx \sin(\pi/|H_{\text{down}}|) = \sin(\pi/3) = \sqrt{3}/2 \approx 0.866$, lying 3.6% above the PDG value 0.836. The deviation falls within the experimental tension between inclusive and exclusive $|V_{cb}|$ determinations, so $\sin(\pi/3)$ is consistent at the $\sim 2\sigma$ level (**A**). Resolving the inclusive/exclusive tension would sharpen this test directly.

9 Discussion

9.1 Comparison with holographic dark energy

The closest prior work to the present cosmological-constant mechanism is the holographic dark-energy programme of Cohen, Kaplan, and Nelson [4] and Hsu [9]. All three share the key structural insight that gravity correlates ultraviolet and infrared scales and thereby depletes the effective number of field-theory modes, yielding $\rho_\Lambda \sim M_{\text{Pl}}^2 H_0^2$. The GTE construction differs in two respects. It fixes the numerical coefficient 9/112 from the CMCA architecture, whereas the holographic bound leaves it free. And it predicts $w = -1$ from the PSC boundary condition, whereas the holographic density tracks the horizon and predicts $w = 0$ — the obstruction that prevents holographic dark energy from producing accelerated expansion [9]. The GTE dark-energy density is a constant of the cosmic evolution by self-containment, not a horizon-tracking quantity, which is precisely why the equation of state is that of a true cosmological constant.

A further distinction from both CKN and Bekenstein–Hawking holography is the *dimension* of holographic encoding. Standard BH holography encodes a 3D volume on its 2D surface, giving mode

suppression $L^2/L^3 = 1/L$ and density scaling $M_{\text{Pl}}^3 H_0$ — still $\sim 10^{61}$ times larger than observed. The three-tape CMCA encodes the 3D spacetime in *one*-dimensional tapes, giving the stronger suppression $3L/L^3 = 3/L^2$ and the observed $M_{\text{Pl}}^2 H_0^2$ scaling (Equation (2)). The CMCA is thus more holographically compressed than a black hole, and this additional compression is precisely the mechanism that bridges BH holography to the correct cosmological-constant magnitude.

The holographic L^1 bound studied here (Section 3.2) may also have implications for particle-mass predictions: the Information Mass Transformer of the GTE verifier [11] currently uses the Bekenstein L^2 bound; upgrading to the tighter CMCA L^1 bound is a direction for follow-on work.

9.2 Holographic non-renormalization versus supersymmetry

Supersymmetry protects the vacuum energy by cancellation between bosonic and fermionic loops, requiring degenerate superpartners — a structure not observed at accessible energies. The holographic non-renormalization of Section 3.6 is different in kind: the loop suppression $3H_0^2/M_{\text{kink}}^2$ is geometric, arising from the boundary mode count $3L$ rather than the bulk count L^3 , and requires no partner spectrum. The $\approx 10^{-42}$ suppression is a property of the holographic architecture, not an algebraic cancellation between species.

9.3 Open problems

Open Problem 1 (Path-integral non-renormalization in interacting sectors). The holographic mode count suppresses sector DOF loops to $\rho_{\text{graded}}^{\text{holo}} \approx -5 \times 10^{-102} \text{ GeV}^4$ (shown above, Section 3.6), confirming that the holographic bound replaces the QFT vacuum-energy computation entirely. In the *pure*- Φ_{MDL} sector all computable loop corrections to Λ vanish identically to all orders: `lambda_all_order_zero_pure_phi` ($\mathbf{A}_{\text{Lean}}$); see also [13] for the full CC derivation. The remaining open question is the *interacting* (non-pure- Φ_{MDL}) sectors: a complete Φ_{MDL} path-integral computation showing that the mode sum is exactly $3L$ (not an ultraviolet-divergent L^3) in the presence of matter couplings would upgrade the holographic non-renormalization result from $\mathbf{A}/\mathbf{D}$ -partial to $\mathbf{A}/\mathbf{D}$.

Open Problem 2 (H_0 from baryon asymmetry — Path D at $\mathbf{A}/\mathbf{D}$; Regime C open). The non-circular route $H_0(T_{\text{CMB}}; \Omega_{\Lambda}, \eta_B, \Omega_{\text{DM}} h^2, G_N)$ gives $H_0 = 67.95 \text{ km/s/Mpc}$ ($\mathbf{A}$ numerical chain) with $T_{\text{CMB}} = 2.7255 \text{ K}$ as the sole external input. The FKTT mechanism derives $\eta_B = 6.109 \times 10^{-10}$ ($+0.15\sigma$ from Planck 2018 CMB+BBN; $\mathbf{A}_{\text{Lean}}$, zero sorry, zero axioms); the asymptotic kink-overlap formula gives 6.36×10^{-10} ($+4.3\sigma$) as an upper bound. The fully first-principles Path D chain — deriving T_{CMB} from the η_B baryogenesis chain with no external input — achieves $\mathbf{A}/\mathbf{D}$ status via `gtb_kink_overlap_certified` ($\mathbf{A}_{\text{Lean}}$). The sole remaining open piece is Regime C: deriving T_{CMB} from the reheating temperature T_{reh} via entropy conservation without presupposing ρ_{Λ} (currently tautological via the radiation-CC entropy balance). This Regime C step is $\mathbf{D}$.

Open Problem 3 (CMB tilt field-theory identification). The algebraic chain giving $n_s = 1 - \ln 2/(2\pi^2)$ is $\mathbf{A}_{\text{Lean}}$. The physical identification of the binary effective field theory with a classical scalar field on a compact S^4 governing the perturbation spectrum is $\mathbf{A}/\mathbf{D}$ and not yet $\mathbf{A}_{\text{Lean}}$.

Open Problem 4 (Wolfenstein A parameter and the Jarlskog residual). The 3.6% gap in $A = \sin(\pi/3)$ accounts for most of the 5% residual in the Jarlskog invariant. A mechanism deriving the A parameter from the Φ_{MDL} Yukawa structure would close both.

Open Problem 5 (Gravitational-wave background and dark-matter relic density). The kink emission of Section 5 produces a stochastic gravitational-wave background from evaporating primordial black holes; its frequency peak and amplitude are not yet computed. (The domain-wall annihilation background is computed in Section 7; the open item here is the PBH kink-emission background.) The relic density of the dark sector is likewise open.

9.4 Falsifiability

Table 4 states the falsifiability profile. Every prediction is sharp and testable; several are within reach of near-future experiments.

GTE predicts cosmic birefringence $\beta = 0^\circ$ exactly (**A/D**): the photon sector occupies the $\mathbb{Z}_7$ -symmetric winding class $w = 0$, decoupling it from the substrate chirality. Current CMB polarization analyses show a 2.7σ hint at $\beta \approx 0.30 \pm 0.11^\circ$ (2021 data); the LiteBIRD satellite (expected 2031) will reach $\sigma(\beta) \approx 0.05^\circ$, constituting a definitive test. A confirmed $|\beta| > 0.15^\circ$ at 5σ would falsify this prediction.

Prediction	Observable	Current bound	How to falsify
$\Omega_\Lambda \in [0.673, 0.690]$	CMB + BAO	0.6889 ± 0.0056	future CMB shifting Ω_Λ outside the range
$n_s = 0.96488$	CMB power spectrum	0.9649 ± 0.0042	any future shift of n_s by > 0.01
$\Sigma m_\nu = 59.4$ meV	CMB-S4 + Euclid	< 120 meV	direct measurement excluding ~ 60 meV
kink emission near $M_{\text{kink}}^{\text{crit}}$	PBH spectrum	none yet	null of a non-thermal bump near 290 MeV
$V_{\text{max}} = 2m^2/49$	primordial GW/PBH	unconstrained	primordial spectra inconsistent with the cutoff
$w_a = 0$ (CPL dark-energy evolution)	w_a (DESI, Euclid)	DESI Y1: $w_a = -0.4 \pm 0.3$ (1.33σ , not yet a challenge)	$ w_a > 0.45$ at 5σ (Euclid $\sigma(w_a) \approx 0.09$)
zero surviving $\mathbb{Z}_7$ wall network	CMB anisotropy, GW background, wall crossing	none observed	any confirmed cosmological domain-wall relic falsifies the defect-cosmology chain of Section 7 (bias-annihilation mechanism + ordering-crossover epoch), not the full framework
cosmic birefringence $\beta = 0^\circ$ (A/D)	CMB polarization rotation	$\beta \approx 0.30 \pm 0.11^\circ$ (2.7σ hint, 2021 data)	$ \beta > 0.15^\circ$ confirmed at 5σ (LiteBIRD, $\sigma(\beta) \approx 0.05^\circ$, 2031)
$\Omega_\Lambda \geq 3\pi/14 = 0.6732$ (carrier floor)	CMB + BAO	Planck 2.80σ above the floor	a confirmed $\Omega_\Lambda < 0.6732$ refutes the floor-half of the bracket
$\Omega_\Lambda < 0.6899$ (census interiority)	future CMB at $\sigma \leq \sigma^* = 3.4 \times 10^{-4}$	Planck $\sigma = 0.0056$	a central value at/above the census endpoint at precision σ^* refutes the capacity-ceiling orientation (one-sided test)

Table 4: Falsifiability profile of the GTE cosmological predictions. The $w_0 = -1$, $w_a = 0$ prediction follows from the PSC boundary condition (**A/D**): dark-energy density is fixed at its epoch-selected value, yielding a true cosmological constant with no CPL evolution. The defect-network row is scoped: a confirmed wall relic falsifies the bias-annihilation chain and the $\mathbb{Z}_7$ ordering-crossover epoch, not the substrate identification or the particle-spectrum derivations.

10 Conclusions

We have assembled the cosmological predictions of the GTE/ Φ_{MDL} framework and derived them from zero-parameter first principles. The cosmological constant is addressed at three levels. Its *value* is fixed by two structurally independent counts — the PSC reflexive-closure route $\Omega_\Lambda = 0.6899$ and

the holographic route $\Omega_\Lambda = 3\pi/14 = 0.6732$ — that bracket the observed 0.6889. The bracketing is itself a theorem: the holographic value prices the MDL-minimal self-description carrier (the lower endpoint) and the PSC value is the capacity ceiling of the same adjudication ledger (the upper endpoint), with the inter-route ordering certified by the GTE-atom inequality $14 \ln(2000/3) > 9\pi^2$; the realized fraction is a definite but non-computable real of degree $\mathbf{0}'$ strictly between them, so the 2.42% spread is the unresolved diagonal capacity, and no computable selection mechanism for the exact value exists. The same orientation excludes every generation count $N \geq 4$ with zero observational input, and together with the PSC Layer-I bound pincers $N_{\text{gen}} = 3$ without Presentation Invariance.

Its *magnitude* follows from the holographic mode count, $\rho_\Lambda = (9/112) M_{\text{Pl}}^2 H_0^2$, with the equation of state $w = -1$ fixed by the PSC boundary condition rather than by dynamics, evading both Weinberg’s no-go theorem and the Cohen–Kaplan–Nelson $w = 0$ obstruction. Its *quantum protection* follows from the same holographic counting, which suppresses one-loop corrections by $\approx 10^{-42}$ relative to the standard field-theory estimate.

The defect-cosmology sector is closed: the $\mathbb{Z}_7$ wall foam that necessarily forms at the ordering crossover $T_G \approx 0.70$ GeV is annihilated within $\sim 10^{-23}$ s by the canonical coupling bias, the universe lands in the vacuum $k = 0$, and the framework predicts zero surviving defect network with $\Omega_{\text{GW}} h^2 \lesssim 3 \times 10^{-47}$ and $\Delta N_{\text{eff}} \approx 0$ — parameter-free and falsifiable.

The CMB scalar spectral index $n_s = 1 - \ln 2/(2\pi^2) = 0.96488$ agrees with Planck 2018 at -0.005σ , certified by fourteen zero-**sorry** theorems. Together with the flatness, horizon, and relic results, this establishes that all four standard inflation motivations are discharged from GTE first principles without an inflationary epoch; a comprehensive treatment appears in [13]. The Newton-constant normalization, the strong-field bound, Hawking kink emission, the neutrino sum, and the CKM CP phase follow from the same substrate. All central results are machine-checked in Lean 4. The remaining open problems — the formal path-integral proof of holographic mode-sum replacement (sector DOF loops are shown to be negligible at $\sim 10^{-102}$ GeV⁴), the derivation of H_0 from the baryon asymmetry, and the field-theory identification of the CMB tilt — are stated precisely, and the predictions are sharply falsifiable.

Unification through the GTE polynomial. Every result in this paper is a consequence of a single combinatorial object: the Generative Triple Evolution polynomial over GF(7), specified in full by 19 bits. From this one polynomial the CMCA architecture is fixed, which determines $N_{\text{spatial}} = 3$, $|\mathbb{Z}_7| = 7$, and $D = 4$, which in turn give $\tau = 3/7$ and the holographic mode count $3L$, which yield $\Omega_\Lambda = 3\pi/14$. The orbit count $|\mathcal{O}_{\text{GTE}}| = 2000/3$ determines $\Omega_\Lambda = 0.6899$ by PSC reflexivity. The same polynomial fixes the $\mathbb{Z}_7$ winding structure, giving n_s , G_N , M_{crit} , Σm_ν , and δ_{CP} . The fact that so many cosmological observables — spanning 60 orders of magnitude in energy scale — all trace back to a single 19-bit specification is perhaps the strongest argument for taking the GTE framework seriously as a candidate foundation for physics.

Acknowledgments

The author thanks the open-source Lean 4 and Mathlib communities, whose libraries underpin the machine certifications reported here.

A Lean Certification Inventory

Table 5 lists the theorems cited in this paper, all in the canonical `ugp-lean` library. Unless noted, each has zero sorry and zero custom axioms.

Table 5: Lean certification inventory.

Theorem	Module	Status	Section
<code>psp_epoch_selection_master</code>	<code>PSCEpochSelection</code>	<code>A_{Lean}</code>	3.3
<code>incompleteness_implies_nonzero_omega_lambda</code>	<code>PSCEpochSelection</code>	<code>A_{Lean}</code>	3.3
<code>omega_lambda_from_d_res</code>	<code>PSCEpochSelection</code>	<code>A/D</code>	3.3
<code>d_res_pos</code>	<code>PSCEpochSelection</code>	<code>A/D</code>	3.3
<code>d_res_determines_omega_lambda</code>	<code>PSCEpochSelection</code>	<code>A_{Lean}</code>	3.3
<code>gauge_spectrum_total</code>	<code>OQ26Arithmetic</code>	<code>A_{Lean}</code>	3.3
<code>lmodel_numerator_coprime_denominator</code>	<code>OQ26Arithmetic</code>	<code>A_{Lean}</code>	3.3
<code>voxel_cc_coefficient</code>	<code>TemporalVoxelCC</code>	<code>A/D</code>	3.2
<code>omega_lambda_from_temporal_voxel</code>	<code>TemporalVoxelCC</code>	<code>A/D</code>	3.4
<code>omega_lambda_via_gorard</code>	<code>TemporalVoxelCC</code>	<code>A_{Lean}</code>	3.4
<code>voxel_coeff_eq_two_c_gorard_tau</code>	<code>TemporalVoxelCC</code>	<code>A_{Lean}</code>	3.2
<code>c_gorard_eq_n_spatial_over_two_dsq</code>	<code>TemporalVoxelCC</code>	<code>A_{Lean}</code>	2.2
<code>holographic_voxel_scaling</code>	<code>TemporalVoxelCC</code>	<code>A/D</code>	3.6
<code>holographic_mode_count_ratio</code>	<code>TemporalVoxelCC</code>	<code>A/D</code>	3.6
<code>holographic_identification_voxel_cc</code>	<code>TemporalVoxelCC</code>	<code>A/D</code>	3.6
<code>holographic_nrt_loop_tree_ratio_h0_independent</code>	<code>TemporalVoxelCC</code>	<code>A/D</code>	3.6
<code>tau_three_sevenths_from_ether</code>	<code>EtherProperTimeRate</code>	<code>A/D</code>	2.2
<code>ether_period_seven</code>	<code>EtherProperTimeRate</code>	<code>A/D</code>	2.2
<code>ether_odd_fire_count</code>	<code>EtherProperTimeRate</code>	<code>A/D</code>	2.2
<code>phimdl_tau_from_ether</code>	<code>PhiMDLProperTimeBridge</code>	<code>A/D</code>	2.2
<code>phimdl_voxel_coeff_from_ether_tau</code>	<code>PhiMDLProperTimeBridge</code>	<code>A/D</code>	2.2
<code>phimdl_omega_lambda_from_ether_dynamics</code>	<code>PhiMDLProperTimeBridge</code>	<code>A/D</code>	2.2
<code>l1_to_l2_tau_bridge</code>	<code>PhiMDLProperTimeBridge</code>	<code>A/D</code>	2.2
<code>n_s_formula</code>	<code>CMBSpectralTilt</code>	<code>A_{Lean}</code>	4
<code>n_s_less_than_one</code>	<code>CMBSpectralTilt</code>	<code>A_{Lean}</code>	4
<code>beta_g_from_classical_rg</code>	<code>CMBSpectralTilt</code>	<code>A_{Lean}</code>	4
<code>beta_g_from_weyl_law</code>	<code>CMBSpectralTilt</code>	<code>A_{Lean}</code>	4
<code>classical_discrete_rg_entropy_rate</code>	<code>CMBSpectralTilt</code>	<code>A_{Lean}</code>	4

Theorem	Module	Status	Section
weyl_log_deriv_at_planck	CMBSpectralTilt	$\mathbf{A}_{\text{Lean}}$	4
weyl_algebraic_miracle	CMBSpectralTilt	$\mathbf{A}_{\text{Lean}}$	4
z2_eft_predicts_cmb_tilt	CMBSpectralTilt	$\mathbf{A}_{\text{Lean}}$	4
z7_superselection_preserved_by_flat_metric	MinimalCoupling	$\mathbf{A}_{\text{Lean}}$	4
kappa_SD_eq_10_13	GorardRationalFormula	$\mathbf{A}_{\text{Lean}}$	5
gorard_chain_catAL_master_bundle	GorardRationalFormula	$\mathbf{A}_{\text{Lean}}$	5
strong_field_uv_bound_catad	PMDLGravityTheorems	$\mathbf{A/D}$	5
z7_potential_max_value	PMDLGravityTheorems	$\mathbf{A}_{\text{Lean}}$	5
kink_bps_mass_formula	PMDLGravityTheorems	$\mathbf{A}_{\text{Lean}}$	5
kink_crit_mass_formula	PMDLGravityTheorems	$\mathbf{A}_{\text{Lean}}$	5
kink_over_hawking_temp_ratio	PMDLGravityTheorems	$\mathbf{A}_{\text{Lean}}$	5
kink_bose_enhanced_at_Mcrit	PMDLGravityTheorems	$\mathbf{A}_{\text{Lean}}$	5
hawking_kink_emission_catad	PMDLGravityTheorems	$\mathbf{A/D}$	5
phimdl_tmunu_vacuum_zero	StressEnergyTensor	$\mathbf{A}_{\text{Lean}}$	3.1
z7_vacuum_energy_mass_independent	NRTVacuumEnergy	$\mathbf{A}_{\text{Lean}}$	3.1
ckm_cp_phase_formula	CKMCPPhase	$\mathbf{A}$	8
cp_phase_reduction_ratio	CKMCPPhase	$\mathbf{A}$	8
ckm_A_parameter_sin2	CKMCPPhase	$\mathbf{A}$	8
omega_lambda_bracket_strict	CCOneJumpResidual	$\mathbf{A}_{\text{Lean}}$	3.5
no_computable_gte_rt_convergence_modulus	CCOneJumpResidual	$\mathbf{A}_{\text{Lean}}$	3.5
sigma_star_falsifiability_horizon	CCOneJumpResidual	$\mathbf{A}_{\text{Lean}}$	3.5
cc_floor_orientation_atom_inequality	NgenBracketOrientation	$\mathbf{A}_{\text{Lean}}$	3.5
ngen_bracket_orientation_flip	NgenBracketOrientation	$\mathbf{A}_{\text{Lean}}$	3.5
ngen_pincer_from_layer1_and_orientation	NgenBracketOrientation	$\mathbf{A}_{\text{Lean}}$	3.5
vcoupling_breaks_z7_shift	ZSevenVacuumSelection	$\mathbf{A}_{\text{Lean}}$	3.1
z7_vacua_chi_kinetic_inequivalent	ZSevenVacuumSelection	$\mathbf{A}_{\text{Lean}}$	7.3
wall_bias_minimum_unique	ZSevenVacuumSelection	$\mathbf{A}_{\text{Lean}}$	7.4
compact_completion_minimum_at_k0	ZSevenVacuumSelection	$\mathbf{A}_{\text{Lean}}$	7.4
scalar_cw_step_inward	ZSevenVacuumSelection	$\mathbf{A}_{\text{Lean}}$	7.4
vacuum_unique_fixed_point_z7	PMDLGravityTheorems	$\mathbf{A}_{\text{Lean}}$	7.4

B Numerical Values and Conversion Factors

- $M_{\text{Pl}} = 1.221 \times 10^{19}$ GeV (Planck mass).

- $H_0 = 67.4 \text{ km/s/Mpc} = 1.44 \times 10^{-42} \text{ GeV}$ (Planck 2018).
- $\rho_{\text{crit}} = 3H_0^2 M_{\text{Pl}}^2 / (8\pi)$ (critical density).
- $m_\tau = 1776.86 \text{ MeV}$; $M_{\text{kink}}^{\text{crit}} = (8/49)m_\tau = 290.10 \text{ MeV}$.
- $g = 49/512$ (fundamental coupling); $M_R = 1.11 \times 10^{13} \text{ GeV}$.
- $N_{\text{spatial}} = 3$; $D = 4$; $|\mathbb{Z}_7| = 7$; $\tau = 3/7$.
- $C_{\text{Gorard}} = 3/32$; $\kappa_{\text{SD}} = 10/13$; CC coefficient = $9/112$.

C GTE Predictions versus Observation

Table 6 collects every quantitative prediction against the corresponding measured value.

Quantity	GTE	Measured	Agreement
Ω_Λ (PSC)	0.6899	0.6889 ± 0.0056	$+0.18\sigma$
Ω_Λ (holographic)	0.6732	0.6889 ± 0.0056	-2.8σ
n_s	0.96488	0.9649 ± 0.0042	-0.005σ
Σm_ν	59.4 meV	$< 120 \text{ meV}$	factor 2 below
δ_{CP}	68.51°	68.53°	0.017%
J	3.02×10^{-5}	3.18×10^{-5}	5%
A	$\sin(\pi/3) = 0.866$	0.836 ± 0.015	within 2σ

Table 6: GTE predictions against current observation. Planck 2018 for Ω_Λ and n_s ; PDG for the CKM quantities.

The ruled-out CMB mechanisms include the $\mathbb{Z}_7$ running route ($n_s(\mathbb{Z}_7) = -15.11\sigma$), which is excluded by the superselection argument of Section 4 in favor of the binary $\mathbb{Z}_2$ route.

Computational Tools

All numerical computations use Python 3.9+ with NumPy for linear algebra and array operations, and SciPy for integration and special functions. Key scripts used in this paper:

- `papers/45_three_tape_cmca/scripts/cc_temporal_voxel_formula.py` — temporal-voxel CC coefficient $\rho_\Lambda = (9/112) M_{\text{Pl}}^2 H_0^2$ and $\Omega_\Lambda = 3\pi/14$.
- `papers/45_three_tape_cmca/scripts/gorard_planck_normalization.py` — Gorard coefficient $C_{\text{Gorard}} = 3/32$ and Newton-constant normalization.
- `papers/18_koide_cyclotomic/scripts/gte_cp_phase_final.py` — CKM CP phase $\delta_{CP} = \pi/2 - 3/8$ and Jarlskog invariant J .
- `papers/21_neutrino_masses/scripts/gte_leptogenesis.py` — leptogenesis efficiency factor K_1 and baryon asymmetry feasibility.
- `papers/21_neutrino_masses/scripts/neutrino_mass_prediction.py` — neutrino mass spectrum and sum Σm_ν .

The defect-cosmology and CC-bracket scripts are co-located with this paper, in its `scripts/` directory:

- `z7_domain_wall_tension_consistency.py` — wall tension σ , ordering crossover T_G , Friedmann domination solve ($z_{\text{dom}} = 832$), anisotropy-bound factors.
- `z7_domain_wall_kibble_foam.py` — Kibble foam statistics, winding inventory, percolation, biased-vs-unbiased relaxation.
- `z7_domain_wall_bias_annihilation.py` — Z_k normalizations, one-loop free-energy splitting brackets, collapse radius, T_{ann} , GW ceiling, ΔN_{eff} .
- `vacuum_selection_rg_improved_direction.py` and `vacuum_selection_rg_defect_numbers.py` — RG-improved Λ_{GTE} -anchored effective potential, $k^* = 0$ selection battery, final defect-cosmology numbers.
- `cc_floor_orientation_arithmetic.py` — the GTE-atom inequality (12) at 50-digit precision, equivalence web, measurement-absence audit.
- `cc_bracket_dres_consistency.py` — bracket interiority, unresolved capacity fraction 2.42%, precision horizon σ^* .

- `cc_two_route_ngen_scan_canonical.py` and `ngen_bracket_orientation_family.py` — canonical two-route N -scan and the 16-variant ansatz family for the $N \geq 4$ orientation exclusion.
- `cc_floor_ledger_decomposition_model.py` and `pmdl_carrier_zero_point_evaluation.py` — carrier-record decomposition certificates (orientation invariance over 2^{16} realized subsets; load-bearing premise nulls) and the exact carrier coefficient chain $N^2/(D^2|\mathbb{Z}_7|) = 9/112$, $\Omega = 3\pi/14$.
- `d_res_left_ce_degree_certificate.py` — left-c.e. monotone approximation, decode-from-value reduction, and convergence-stage certificates for the realized-ledger classification.

The machine-checked formalisation is written in Lean 4 with Mathlib. All theorems cited as $\mathbf{A}_{\text{Lean}}$ carry zero `sorry` and zero custom axioms beyond Lean’s standard foundations; named physics axioms are explicitly disclosed wherever used. The full theorem inventory is in Appendix A.

Full reproduction instructions are provided in `REPRODUCE.md` co-located with this paper’s source.

AI coding assistants were used for LaTeX formatting, coding assistance, and adversarial review during manuscript preparation. All mathematical derivations, numerical computations, and Lean 4 certifications were independently verified by deterministic scripts and the Lean 4 kernel.

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